

When Answer Agreement Fails: Retrieval-State Lock-In in Retrieval-Augmented Generation

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Abstract

Many black-box uncertainty methods used with retrieval-augmented generation (RAG) read agreement among sampled answers as confidence. That reasoning fails under **retrieval-state lock-in**: repeated samples see the same defective retrieval state, empty or wrong, and the answer settles around the error. To diagnose the failure, the analysis separates three objects that a single confidence score conflates: answer-state uncertainty (variation across sampled answers), evidence-state uncertainty (whether retrieved passages support the answer), and retrieval-state diagnostics (whether the graph retrieval state supplies answer-relevant support). Across six question-answering (QA) snapshots in an ontology-guided knowledge-graph RAG (KG-RAG) system, answer-level uncertainty misses many errors even under the deployable policies: at the operational five-sample budget ($N = 5$), 42% of KG-RAG and 59% of dense-retrieval errors show zero answer dispersion. The collapse is sharpest in a strict graph-only stress test (a mechanism probe, not a deployment policy or one-component ablation); there, an answer-dispersion score ranks clinical errors worse than chance (AUROC 0.233). A route decomposition splits the silent mass into an empty retrieval state (*absence*) and wrong-but-coherent neighbourhoods on bridge questions (*presence*); the latter remains open. The recommendation follows directly: report the three families separately, and treat an answer as low-risk only when all three agree, a conjunctive audit rule that selects a small, high-precision subset. The rule is strongest on the clinical calibration domain and weaker on open-domain multi-hop QA. The point is diagnostic, not a claim that KG-RAG is more accurate than dense retrieval: confidence in RAG is object-specific, and exposing the retrieval state makes some confidence failures easier to diagnose.

1 Introduction

Whether a RAG system can be trusted depends on more than the generated answer. Retrieval relevance, evidence support, and usable provenance all matter (Huang et al., 2024; Ni et al., 2025). Yet many black-box uncertainty estimators used with RAG still begin at the answer surface. Semantic entropy (Farquhar et al., 2024; Kuhn et al., 2023), SelfCheck-GPT (Manakul et al., 2023), and their RAG-integrated variants (Jiang et al., 2023; Su et al., 2024; Zimmerman et al., 2024) ask, in different forms, whether sampled answers agree. Agreement is then used as a confidence signal, disagreement as uncertainty. That assumption fails when retrieval repeatedly concentrates on the same defective state. The answer is stable because the error is stable. The problem is not that responses depend on retrieval; that is expected. The problem is that answer agreement loses diagnostic value when the retrieval state stops varying. The mechanism is plain: a fixed context yields a fixed answer distribution. The contribution is to measure its reach, namely how often silent errors compatible with lock-in arise under deployable policies and which diagnostic signals still expose them.

Retrieval-state lock-in. I call this failure mode **retrieval-state lock-in**: repeated samples condition on the same de-

fective retrieval state and therefore produce the same wrong answer. Two variants matter for diagnosis. In *absence* lock-in, retrieval repeatedly returns no usable graph state, so the model answers from parametric memory. In *presence* lock-in, retrieval repeatedly returns a coherent but wrong graph neighbourhood. In both cases, the shared ingredient is stability, not correctness.

One presence-lock-in case asks which pharaoh the temple behind the Osireion honours (gold: *Seti I*): the retriever anchors to *Ramesses II*, all five samples repeat that wrong name with zero answer-level entropy around a coherent graph trace. The trace in Figure 1 shows why only the contradicting passages raise the alarm. Unlike ordinary noisy retrieval, where contexts vary, or the sample-consistency view of hallucination, where confabulations diverge across samples (Manakul et al., 2023; Farquhar et al., 2024), the error here is stable.

Two definitions fix the scope.

Definition 1 (Silent error) A wrong answer with zero observed answer-state dispersion under the sampling budget (all N samples in one semantic cluster, $SD-UQ$ at its floor).

Definition 2 (Retrieval-state lock-in) A silent error for which the repeated samples condition on the same defective retrieval state, either empty (*absence*) or a coherent but

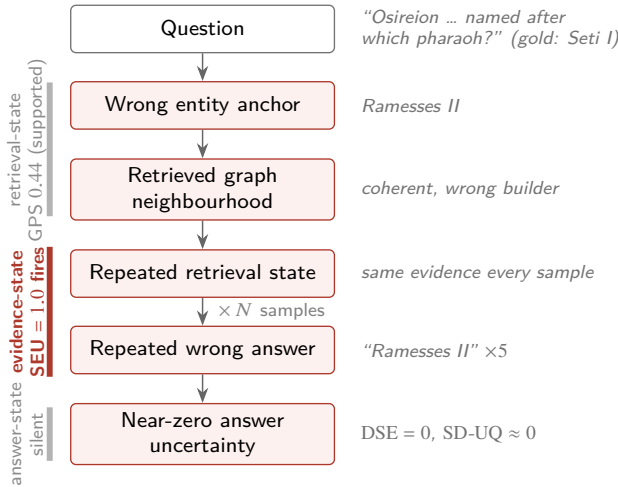


Figure 1: Worked presence-lock-in trace, instantiated with the HotpotQA Osireion case also listed in Appendix Table 30. The retriever anchors to a strongly associated but wrong entity, the resulting graph neighbourhood is internally coherent, every sample receives the same evidence, and the repeated wrong answer presents near-zero answer-state uncertainty. The left rail shows where each diagnostic family attaches: the answer-state family is silent by construction, GPS remains low because the wrong answer is reachable in the retrieved graph, and only the evidence-state signal fires—the retrieved passages contradict the repeated answer. This traces the *presence* variant; in the *absence* variant (Section 7.4) the chain instead repeats an empty retrieval state, and the warning moves to the retrieval-state abstention observable.

wrong neighbourhood (presence).

Silence is a necessary symptom of lock-in, not proof of it: a silent error can also arise from parametric overconfidence, benchmark ambiguity, or answer-normalisation artefacts. The analysis reports those symptom counts and uses route logs to separate the cases that can be diagnosed from saved traces.

Lock-in is easier to diagnose in knowledge-graph-augmented RAG (KG-RAG), where a symbolic retriever can repeatedly anchor to the same entity and traverse the same relation path. Knowledge graphs can help on multi-hop or provenance-sensitive tasks (Edge et al., 2024; Gutiérrez et al., 2024; Hu et al., 2025; Shen et al., 2025; Zhu et al., 2025a; Wu et al., 2025). Recent comparisons, however, find graph retrieval task-dependent and often complementary to flat dense retrieval rather than universally superior (Han et al., 2025; Xiang et al., 2025; Chen et al., 2025; Peng et al., 2024). That same symbolic structure can sharpen the failure mode, because entity-first retrieval may keep returning the same wrong neighbourhood.

The symptom is not limited to graphs. Any system that repeatedly returns a concentrated evidence set can produce stable wrong answers, but KG-RAG is useful here because its retrieval state is visible: matched entities, triples, paths, and anchors make the mechanism observable rather than

merely suspected. The experiments use OntoGraphRAG,¹ an ontology-guided KG-RAG framework with entity-first routing, dense fallback, provenance-aware evidence assembly, and graph-state logging, as the experimental setting in which to measure and characterise lock-in.

Three distinct uncertainty objects. The central claim is that RAG confidence has three distinct objects:

1. **Answer-state uncertainty** measures variation across sampled answers.
2. **Evidence-state uncertainty** measures whether the retrieved passages locally support the generated answer.
3. **Retrieval-state uncertainty** measures whether the retrieval state itself supplies graph support.

These families are not causally independent; they are useful because each attaches to a different object in the pipeline: the answer, the evidence, and the retrieval state. The triad is used throughout.² Each also maps to a practical trust question for deployed systems: calibrated confidence, faithfulness to retrieved evidence, and auditable provenance.

Contributions. First, the analysis **measures the silent-error footprint compatible with retrieval-state lock-in** under deployable policies, not only in strict stress tests: 42% of adaptive KG errors and 59% of dense-retrieval errors carry zero observed answer-state uncertainty at the operational $N=5$ sampling budget, so a score using only within-question answer dispersion has no disagreement signal to exploit: a family-level blind spot, not a stress-test artefact. Second, it introduces a **diagnostic decomposition** that separates answer-state, evidence-state, and retrieval-state uncertainty. Within the answer-state family, it also introduces SD-UQ, a low-cost question-conditioned embedding-dispersion score that is competitive in these low-sample RAG runs, while showing that even stronger answer-state scores cannot diagnose lock-in alone. The answer-state and evidence-state families are implemented directly; the retrieval-state family is represented by a conditional graph-support diagnostic (GPS, soft-linked and depth-matched, reported in risk orientation: lower means stronger graph support). Their representative scores are weakly correlated in practice (pooled Spearman $\rho = 0.03$ between answer- and evidence-state), so they carry complementary signal rather than restating one scalar. Third, it provides **stress-test evidence**: a strict graph-only probe drives answer-state AUROC down to 0.233 (−0.52 versus adaptive KG under paired bootstrap), and route logs localise much of the silent error mass to repeated empty retrieval states. Fourth, it reports a **controlled empirical study** across five QA families and distills the decomposition into a **conjunctive low-risk audit rule**: treating an answer

¹<https://github.com/julka01/OntoGraphRAG>

²Some figures and artefact names use the aliases *output*, *grounding*, and *structural*.

as low-risk only when all three families concur selects a low-risk subset at 91.9% pooled accuracy against a 69.7% base rate, and at 100% precision under the automated judge on the clinical target domain at 22% coverage (Section 7.8).

To separate deployment behaviour from the stress regime, the experiments compare dense RAG, adaptive KG-RAG, and a strict graph-only KG stress test across clinical, biomedical, and open-domain multi-hop QA snapshots. The adaptive setting is the practical policy. The strict graph-only setting is a mechanism probe, not a proposed retriever. The results are mixed by design: adaptive KG retrieval is close to dense retrieval accuracy on several snapshots (e.g., 0.946 vs. 0.950 on RealMedQA) but does not universally improve it. That is the condition under which the diagnostic contribution becomes precise: when accuracy parity holds, what matters is what the retrieval layer makes visible.

These pieces lead to the audit rule evaluated in Section 7.8.

Scope. Three boundaries constrain the claims below. (1) The absence/presence route decomposition depends on the symbolic retrieval state that OntoGraphRAG exposes; the silent-error phenomenon itself is not graph-specific. (2) The strict graph-only stress test changes several pipeline components at once, so it bounds a regime rather than isolating a single causal variable. (3) The fixed-subset evaluation estimates diagnostic behaviour, not lock-in prevalence in deployed systems.

2 Related Work

2.1 Answer-State Uncertainty for LLMs

A large part of LLM uncertainty estimation operates in output space. Semantic entropy clusters sampled responses into meaning-equivalent groups and computes entropy over the resulting distribution (Kuhn et al., 2023; Farquhar et al., 2024). Related methods simplify or approximate that idea: SEP predicts semantic entropy from hidden states (Kossen et al., 2025), SelfCheckGPT uses contradiction across samples (Manakul et al., 2023), and P(TRUE) asks the model to verbalise its own confidence (Kadavath et al., 2022). The same answer-agreement intuition underlies self-consistency decoding, where agreement across sampled reasoning paths is read as a correctness signal (Wang et al., 2023). SEP is relevant but requires white-box hidden-state access, which the present black-box setting does not provide. More recent white-box methods such as induction-aware entropy gating (Bazarova et al., 2026), which modulates entropy by induction-head activity, are likewise out of scope here but orthogonal: such a mechanistic signal could be fused with retrieval-state diagnostics rather than competing with them.

A second line of work uses response-embedding geometry rather than hard semantic clustering. Kernel Language Entropy measures von Neumann entropy on a similarity ker-

nel (Nikitin et al., 2024); Semantic Density and Semantic Volume quantify concentration and dispersion in embedding space (Qiu and Mikkilainen, 2024; Li et al., 2025b); and Walha et al. (2025) decompose representation uncertainty with a spectral formulation. These methods are the closest answer-state baselines for the family studied here. Their limitation for the present setting is one of scope, not quality: they were developed for answer variation itself, not for regimes where retrieval may stabilise around an explicit graph trace.

2.2 Uncertainty in RAG

RAG uses uncertainty in several ways. Adaptive retrieval methods use confidence or entropy to decide when retrieval should occur (Jiang et al., 2023; Su et al., 2024; Yao et al., 2025; Liu et al., 2025; Zubkova et al., 2025), while others import standalone LLM uncertainty estimators as retrieval triggers or answer-confidence scores (Moskvoretskii et al., 2025). This connects to selective prediction and abstention (Geifman and El-Yaniv, 2017; Tomani et al., 2024), conformal NLP (Campos et al., 2024), and RAG trustworthiness work that treats reliability as a pipeline property rather than a generator property (Huang et al., 2024; Ni et al., 2025). Conformal methods derive a formal coverage guarantee from a single nonconformity score; the conjunctive audit rule introduced here instead requires three signal families to agree and targets auditable abstention rather than formal coverage calibration. This literature shows that uncertainty is already operational in RAG, but many deployment-facing designs still reduce it to one routing or abstention scalar rather than reporting answer, evidence, and retrieval-state behaviour separately. These answer-side gates trigger retrieval from sampled-answer uncertainty, the very signal that lock-in silences, so they compose with rather than replace the evidence- and retrieval-state diagnostics studied here. A complementary line predicts evidence quality directly: Perez-Beltrachini and Lapata (2025) train a lightweight passage-utility model for a target QA system, approximating sampling-based uncertainty without sampling, a learned counterpart to the evidence-state signal whose head-to-head against SEU is a logical next comparison. A neighbouring self-reflective line asks the model itself to decide when to retrieve, critique, or abstain, often through learned control tokens, fine-tuning, or model-specific prompting policies. Self-RAG trains reflection tokens for retrieve/generate/critique decisions; corrective RAG evaluates retrieved document quality and triggers corrective retrieval (Asai et al., 2024; Yan et al., 2024). Self-RAG and corrective RAG need white-box access for fine-tuning or control tokens; the present design instead emphasises black-box diagnostics, because clinical and biomedical deployments often sit behind hosted models, locked APIs, or governance constraints that rule out hidden-state probes and task-specific fine-tuning.

Several recent papers move closer to the present per-

spective. An axiomatic analysis shows that standard uncertainty estimators cannot reliably assess correctness in RAG and proposes a calibration framework to address it (Soudani et al., 2025a); the present study is the empirical counterpart, locating where answer-state agreement collapses and which retrieval- and evidence-state signals still respond. FRANQ separates factuality from faithfulness to retrieved context, but it operates on flat-text RAG and does not expose three-way graph-side decomposition (Fadeeva et al., 2025). SURE-RAG scores evidence-set sufficiency for selective answering (Qiu et al., 2026), the evidence-state question asked well, but not whether retrieval itself has concentrated around a wrong symbolic trace. R2C perturbs the retrieval-reasoning loop so that dispersion under paraphrases, rethinking, and validation reflects both retriever and generator uncertainty (Soudani et al., 2025b). It is complementary: R2C measures movement under perturbation, while lock-in concerns unperturbed retrieval that barely moves at all. This also connects to retrieval-stability and context-selection work more broadly. Long-context RAG and context-balancing methods ask which spans should be kept or loaded (Li et al., 2024; Voloshyn, 2026); noise-aware calibration studies how irrelevant retrieved context affects confidence (Liu et al., 2026a); and retriever benchmarks study how much information the retrieved set carries (Zheng et al., 2026). Those papers establish that retrieval composition matters for reliability. The present claim is narrower: when repeated samples see the same defective state, answer variance can collapse even if the generator is sampled. Ca2KG studies overconfidence in KG-RAG through counterfactual prompting, but it does not report a structured multi-family diagnostic pipeline (Ren et al., 2026). BRINK shows that KG-RAG systems can fall back on parametric memory when graphs are incomplete (Zhou et al., 2026); here the absence/presence split is retrieval-time and is separated empirically in Section 7.4. Table 1 summarises the positioning against these five closest neighbours. A direct empirical head-to-head against them is out of scope here: FRANQ, SURE-RAG, and R2C target flat-text RAG and are reported on different corpora, retrievers, and answer formats, so a like-for-like comparison would mean reimplementing each inside the KG-RAG pipeline. The contribution is the graph-side three-family decomposition and the lock-in diagnosis, which is orthogonal to and composable with those evidence- and perturbation-based scores; a controlled head-to-head, especially against learned passage-utility predictors, is the natural next study.

Noise-aware calibration and long-context RAG work show that retrieval can create uncertainty beyond decoding alone (Liu et al., 2026a; Li et al., 2024), and evaluation frameworks such as RAGAs and ARES foreground context relevance and faithfulness alongside answer quality (Es et al., 2024; Saad-Falcon et al., 2024). RAG evaluation has separated signals before; the narrower contribution here is to connect that separation to explicit graph retrieval state, where answer agreement and evidence support can decou-

Table 1: Positioning against the closest related work (citations in the text above). ✓: a direct object of study; ~: partial or indirect (e.g., perturbation-based rather than state-inspecting); −: not addressed. *Trace* denotes whether the symbolic retrieval state (anchors, paths, triples) is exposed for inspection; *lock-in* denotes whether the concentrated-wrong-retrieval regime itself is studied. Among these five recent neighbours, the present study is the only one marked across all five columns.

	Answer state	Evidence state	Retrieval state	Trace exposed	Lock-in studied
FRANQ	✓	✓	−	−	−
SURE-RAG	−	✓	~	−	−
R2C	✓	~	~	−	−
Ca2KG	✓	−	~	~	−
BRINK	−	−	~	~	−
This paper	✓	✓	✓	✓	✓

ple with measurable consequences for uncertainty quality. Several of these evaluator families are already represented inside the eight-metric suite benchmarked here, semantic entropy and SelfCheckGPT (sampling and cross-sample consistency), a perturbation-sensitivity score (SRE-UQ), a calibration proxy (P(True)), and a RAGAs-style answer-support score (SEU), so the comparison is in part internal rather than absent. What remains for a dedicated head-to-head is the learned and full-pipeline systems (passage-utility predictors, FRANQ, SURE-RAG, R2C, and ARES-style frameworks), each of which targets flat-text RAG and would have to be reimplemented inside the KG-RAG pipeline to compare like for like. This also relates to internal-external conflict, where retrieved evidence supports one answer while the model prior favours another. ClashEval and recent conflict-aware RAG systems model that tension through source-aware synthesis, fact-level conflict modelling, or KG-mediated reconciliation (Wu et al., 2024; Wang et al., 2025; Zhang et al., 2025; Liu et al., 2026b). Recent attribution work makes a related distinction between a citation that is correctly supportive and one the model relied upon (Wallat et al., 2025). The present study asks which signals remain useful when retrieval is visible and partially stabilised.

2.3 KG-RAG and Graph-Side Uncertainty

Graph-augmented retrieval has become a major alternative to flat-text RAG. Systems such as RoG, Think-on-Graph, HippoRAG, SubgraphRAG, and GraphRAG use graphs to make multi-hop reasoning more explicit, decomposable, and auditable (Luo et al., 2024; Sun et al., 2024; Ma et al., 2025; Gutiérrez et al., 2024; Li et al., 2025a; Edge et al., 2024). Their benefits tend to appear on tasks requiring multi-hop retrieval, bridge preservation, networked evidence organisation, or stronger provenance (Edge et al., 2024; Gutiérrez et al., 2024; Shen et al., 2025; Hu et al., 2025; Zhu et al., 2025a; Wu et al., 2025). Topology-aware retrieval makes a related point from the retrieval side: texts can be selected by graph proximity and structural role, not only lexical or

embedding similarity (Wang et al., 2024). This study treats the graph as a source-linked summary of passages: triples, paths, and relation anchors matter because they trace back to the text that licensed them. Comparative work correspondingly suggests that graph and dense retrievers are often complementary rather than globally ordered (Han et al., 2025; Xiang et al., 2025; Peng et al., 2024; Chen et al., 2025).

What remains comparatively underdeveloped is graph-side confidence analysis: scoring whether the retrieved graph state itself, its matched entities, traversal paths, and relation anchors, supports the generated answer, rather than checking answer variation or text-evidence consistency. Ca2KG studies overconfidence in KG-RAG through counterfactual prompting (Ren et al., 2026), and BRINK shows that KG-RAG systems can lean on parametric memory when retrieved graphs are incomplete (Zhou et al., 2026). A separate literature models uncertainty inside knowledge graphs themselves, for example with confidence-weighted triples or distribution shift in KG embeddings (Takahashi et al., 2026; Lee, 2025; Zhu et al., 2025b). The diagnostic question is how to score answer variation, evidence support, and graph support once a hybrid KG-RAG system exposes all three.

3 Background and Problem Formulation

3.1 Sampling-Based Uncertainty in RAG

Let q be a question, c a retrieved context, and r a model response. Sampling-based uncertainty estimators operate on repeated draws from $p(r \mid q)$. In RAG, however, the response distribution is mediated by retrieval:

$$p(r \mid q) = \sum_{c \in C(q)} p(r \mid q, c) p(c \mid q), \quad (1)$$

where $C(q)$ is the finite set of candidate contexts the corpus and retrieval policy make available for question q . In words, the response distribution is a mixture over the contexts the retriever might return, weighted by their retrieval probability. Methods such as semantic entropy therefore reflect both decoder stochasticity and whatever variability retrieval introduces. In vanilla RAG this can be useful: different top- k chunks yield different answers when evidence is weak or unstable.

3.2 Why KG-RAG Changes the Interpretation

KG-RAG changes how that disagreement should be read. Vanilla RAG returns a ranked list of passages; KG-RAG can return a structured state whose elements have names and types, encoding claims as triples (subject, relation, object). Anchors, relation labels, triples, paths, and the passages that license them are checkable intermediate objects, not merely

Observed object	Signal family	Diagnostic question
Answer samples	Answer-state uncertainty	Do repeated generations disagree, or has the answer surface become artificially stable?
Retrieved passages	Evidence-state uncertainty	Does the retrieved text locally entail, contradict, or fail to support the generated answer?
Graph trace	Retrieval-state uncertainty	Which anchors, typed triples, and paths made the answer reachable, and where does the graph abstain?

The families are not independent causes of correctness. They are separately logged views of the same KG-RAG run: generated text, retrieved evidence, and symbolic retrieval state.

Table 2: Uncertainty is treated as an audit problem over pipeline objects. Answer-state metrics see only sampled answers. Evidence-state metrics compare the answer with retrieved passages. Retrieval-state diagnostics inspect the graph trace itself: anchors, subject–relation–object triples, paths, and abstentions. This is why the families are reported separately. This table defines the objects; Table 28 states how each family behaves across retrieval regimes, and Table 6 maps their joint outcomes into a 2×2 failure taxonomy.

more prompt text. In the entity-first regime studied here, retrieval is mediated by entity matching and graph expansion over a fixed graph $\mathcal{G}$. When that process is highly stabilised, one question-conditioned context can dominate:

$$c^* = \arg \max_{c \in C(q)} p(c \mid q). \quad (2)$$

As the residual mass $1 - p(c^* \mid q)$ shrinks, the marginal in Equation (1) is dominated by a single term,

$$p(r \mid q) \approx p(r \mid q, c^*), \quad (3)$$

and repeated samples mostly expose decoder variability under fixed evidence (Appendix A gives the corresponding total-variation bound for the small-residual regime). Equation (3) is a stylised limit, not a claim about KG-RAG as a category, and the experiments do not estimate $p(c \mid q)$ directly. It formalises the regime of interest: once retrieval stabilises, low answer variance can mean either good evidence or the same retrieval-side mistake on every sample.

3.3 A Three-Family Analysis Frame for KG-RAG

“Uncertainty in KG-RAG” is not a single quantity: entity anchoring, graph expansion, evidence assembly, and generation can fail for different reasons. The analysis tracks the

three families against three observable objects: answer, evidence, and retrieval state (Table 2). The factorisation below is a bookkeeping device for these observable stages, not a distinction implied by probability theory alone. Let s denote the graph retrieval state (anchors, paths, and subgraph), c the textual evidence assembled from it, and r the answer. The pipeline is then

$$p(r, c, s | q) = p(s | q) p(c | q, s) p(r | q, c, s). \quad (4)$$

Answer-state uncertainty primarily probes variability in $p(r | q, c, s)$; evidence-state measures probe whether the produced evidence c is locally consistent with the answer r ; retrieval-state measures probe whether the retrieval state s supplies coherent graph support in the first place. These are consistency diagnostics over the retrieved state, not direct correctness tests, and the split is not a claim of independence: evidence-state signals depend on retrieval, retrieval-state scores depend on entity extraction and graph coverage, and answer variation depends on retrieval stability. It is a reporting partition, not a causal ontology; a system may produce low-variance answers from the wrong subgraph, retrieve a plausible path whose passages do not entail the answer, or retrieve good evidence while the answer remains unstable.

4 System Description

The experimental platform is OntoGraphRAG, an open-source framework that exposes both vanilla RAG and KG-RAG under a shared interface. The two systems use the same corpora, embeddings, and generation model; they differ only in retrieval and evidence organisation.

4.1 OntoGraphRAG Overview

OntoGraphRAG is designed around a simple principle: graph retrieval should add structure without discarding dense evidence (Figure 2). The graph makes retrieval commitments explicit: recognised entities, accepted relations, triples, paths, and licensing passages. During KG construction, passages are converted into entity and relation nodes with ontology-guided typing, anchor grounding, confidence scores, provenance links, and contradiction flags when a schema is available. During retrieval, the system first tries to route the question through matched entities and local graph neighbourhoods. When the anchor is weak, it uses dense retrieval and retriever-first graph expansion as fallback. The resulting context contains both textual passages and explicit graph paths, so the answer can be evaluated at three levels: the answer state, the evidence state, and the retrieval state. OntoGraphRAG logs anchors, paths, and supporting passages with the final answer, so the provenance trace is part of the retrieval output, not a private prompt construction detail.

4.2 Vanilla RAG

Vanilla RAG performs direct vector retrieval over chunk embeddings. For each question, it retrieves the top- k chunks above a similarity threshold and optionally appends adjacent chunks to capture answers split across chunk boundaries. The LLM therefore receives flat text only. It has no explicit entities, relations, or multi-hop paths, and thus no graph-side object on which to define retrieval-state diagnostics.

4.3 KG-RAG

KG-RAG replaces flat retrieval with a routed hybrid pipeline rather than a single graph walk.

Stage 1: seed selection. The preferred path is entity-first anchoring: the system identifies seed entities with symbolic matching and embedding lookup over short mentions. If no reliable anchor is found, the system does not immediately reduce to vector-only retrieval; instead it can route to a retriever-first graph pass.

Stage 2: local graph expansion and scoring. Starting from the seed entities, the system traverses the graph up to a dataset-specific hop limit, producing an explicit retrieval state of linked chunks, entities, relations, and readable traversal paths. Traversal is provenance-aware: on question-bundle datasets, paths and edges are restricted to the current question’s local evidence rather than allowed to borrow support from other questions in the same dataset graph. Chunks are scored by hop distance and local diffusion over the retrieved entity subgraph.

Stage 3: retriever-first graph expansion when anchoring is weak. When entity-first retrieval is weak or empty, the system first retrieves dense text chunks, extracts their linked entities, expands the graph from those passage-derived seeds, and re-scores the resulting context with a combined vector-plus-graph signal.

Stage 4: evidence organisation. The final prompt is not a flat mixture of chunks and graph paths. Instead, retrieved graph paths are grouped with overlapping supporting passages into explicit reasoning chains, with remaining passages listed separately as additional evidence. The generator sees both structured multi-hop support and the local text that grounds each chain.

Diagnostic role. OntoGraphRAG is useful here because it makes the retrieval state explicit rather than tacit. The graph is not treated as an oracle; it is an auditable substrate whose triples and paths may be helpful or wrong, but are diagnostically richer than an unstructured list of nearest-neighbour chunks. Because the system logs which entities

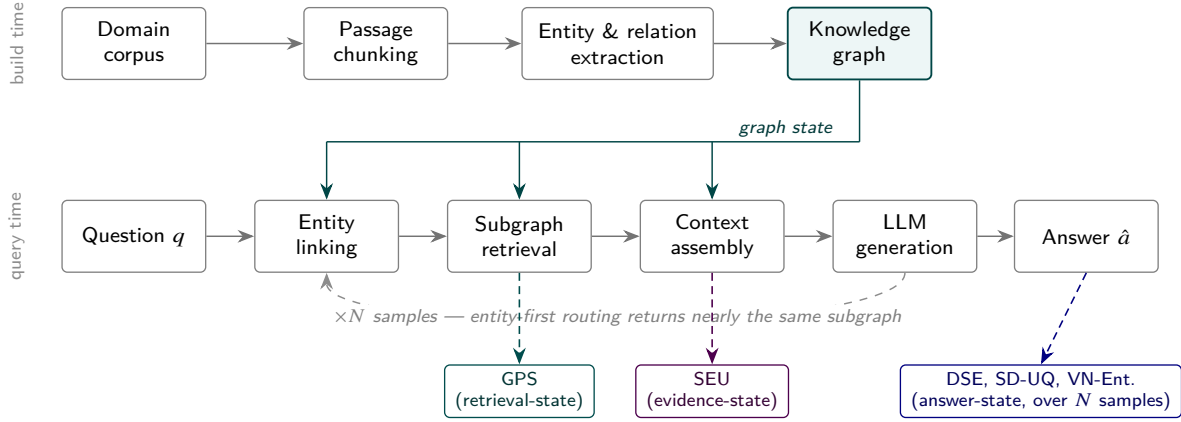


Figure 2: OntoGraphRAG pipeline with the three measurement taps. *Build time* (top): a domain corpus is chunked and converted via ontology-guided extraction into a typed knowledge graph with passage-level provenance. *Query time* (bottom): the question is linked to seed entities, and the graph state conditions retrieval at three points—entity linking, subgraph traversal, and context assembly—before the LLM produces the final answer. The dashed probes mark where each diagnostic family attaches: GPS scores graph support in risk orientation, SEU scores the assembled evidence against the answer, and the answer-state estimators see only the N sampled answers. The sampling loop is the lock-in mechanism: each of the N samples re-enters retrieval, and entity-first routing returns nearly the same subgraph, so answer-state uncertainty mostly reflects decoder noise under almost-fixed evidence (Section 5).

were matched, which paths were traversed, and which route was taken, the retrieval state is an explicit, queryable object rather than an internal variable.

Retrieval stabilisation. The diagnostic question arises when the retrieval state is stable across repeated calls; retriever-first expansion, vector fallback, and iterative decomposition can weaken that stabilisation when the graph anchor is brittle, so the experiments compare diagnostics across retrieval regimes. Entity-first retrieval can be especially prone to stabilising: once the system commits to an anchor entity, the subsequent graph expansion is largely determined, so the same neighbourhood, right or wrong, tends to recur across samples.

5 Uncertainty Measures

Eight diagnostic measures are evaluated across three observable objects in a QA episode: sampled answers, retrieved graph state, and retrieved evidence (Table 3). They are not eight independent estimates of one hidden confidence variable. Within-family dependencies are explicit: the $P(\text{True})$ -style proxy is monotone in DSE under black-box sampling, and VN-Entropy and SD-UQ are both geometric statistics of the same response matrix. The headline tables report family representatives rather than treating all eight rows as separate evidence. The answer-state scores mostly come from standalone LLM uncertainty (Kuhn et al., 2023; Manakul et al., 2023; Kadavath et al., 2022; Moskvoretiskii et al., 2025); the retrieval- and evidence-state scores ask whether the generated answer is reachable in the graph and supported by the retrieved text. Let $\mathbf{r} = \{r_1, \dots, r_N\}$ denote N responses sampled from the LLM for a question q with retrieved con-

text c . Let $\mathbf{v}_i \in \mathbb{R}^d$ denote the ℓ_2 -normalised embedding of r_i , and $\mathbf{V} = [\mathbf{v}_1 \mid \dots \mid \mathbf{v}_N]^\top \in \mathbb{R}^{N \times d}$ the stacked response embedding matrix. All estimators return a non-negative scalar with higher values indicating greater uncertainty. The retrieval-state diagnostic GPS and the evidence-state estimator SEU are bounded in $[0, 1]$; DSE and VN-Entropy have maximum $\log N$; SRE-UQ and SD-UQ are non-negative and unbounded. Because the experiments use only $N = 5$ samples, the answer-state estimators are low-sample operational diagnostics, not high-precision distribution estimates.

5.1 Answer-State Uncertainty Estimators

Semantic entropy and DSE. Kuhn et al. (2023) cluster sampled responses into meaning-equivalent sets $\mathcal{C} = \{C_1, \dots, C_K\}$ using bidirectional NLI entailment. The API setting exposes no reliable per-sample likelihoods, so the reported score is the count-weighted black-box variant,

$$\text{DSE}(\mathbf{r}) = - \sum_{k=1}^K \frac{|C_k|}{N} \log \frac{|C_k|}{N}. \quad (5)$$

DSE is semantic entropy with uniform sample weights; likelihood-weighted SE is therefore not reported separately.

$P(\text{True})$. $P(\text{True})$ (Kadavath et al., 2022) is implemented as a black-box cluster-agreement proxy: the fraction of samples in the same semantic cluster as the first response ($P(\text{True})_{\text{proxy}} = |\{i : \text{cl}(r_i) = \text{cl}(r_1)\}|/N$); uncertainty is $1 - P(\text{True})_{\text{proxy}}$. It is monotone in DSE here and is not treated as independent evidence.

Table 3: Taxonomy of the eight uncertainty measures. The first six rows are answer-state estimators; the last two instantiate retrieval-state and evidence-state diagnostics. Measured per-question costs are reported in Appendix Table 20. †: metric or KG-RAG operationalisation defined here. ‡: prior estimator first benchmarked here for KG-RAG.

Family	Measure	Formal sketch	Required inputs	Cost at inference
Entropy	DSE	$-\sum_k \hat{p}_k \log \hat{p}_k$, with $\hat{p}_k = C_k /N$	sampled answers, NLI	N samples; $O(N^2)$ NLI calls
Calibration	P(True)	$1 - \{i : \text{cl}(r_i) = \text{cl}(r_1)\} /N$; black-box P(True)-style proxy	sampled answers, NLI	N samples; shares DSE clustering
Similarity	SelfCheckGPT	$\frac{1}{ \mathcal{P} } \sum_{(i,j) \in \mathcal{P}} [\text{NLI}(r_i, r_j) = \text{CONTRADICTION}]$	sampled answers, NLI	N samples; $O(N^2)$ NLI calls
Perturbation	SRE-UQ	$\frac{1}{M} \sum_{i \in \mathcal{T}} \Delta_i $; perturbation sensitivity of the kernel mean embedding	sampled answer embeddings	N samples; embeddings only
Geometric	VN-Entropy‡	$-\text{tr}(\rho \log \rho)$, with $\rho = VV^\top/N$	sampled answer embeddings	N samples; embeddings only
	SD-UQ†	$\exp(\frac{1}{k} \sum_{i=1}^k \log(\eta_i + \varepsilon))$ on question-orthogonal residuals	question embedding, sampled answer embeddings	N samples; embeddings only
Retrieval-state	GPS†	$1 - \frac{\sum_{e: \text{reach}_e} w_e \gamma^{ L_e - \hat{L}(q) }}{\sum_e w_e}$; soft-linked, depth-matched answer-entity support	retrieved KG, entity embeddings, linked question/answer entities	1 sample; per-entity graph queries, no LLM calls
Evidence-state	SEU†	$\frac{1 - (n_E - n_C)/K}{2}$; entailment-contradiction deficit over retrieved chunks	retrieved chunks, answer, NLI	1 sample; K NLI calls

SelfCheckGPT. Manakul et al. (2023) measure the fraction of response pairs flagged as contradictions by NLI:

$$\text{SCG} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} [\text{NLI}(r_i, r_j) = \text{contradiction}], \quad (6)$$

where $\mathcal{P}$ is the set of ordered response pairs. For these short-answer tasks, the implementation uses whole-response NLI.

SRE-UQ. Vipulanandan et al. (2026) measure perturbation sensitivity of the response embedding distribution. Let Δ_i denote the first-order perturbation score of response embedding ϕ_i around the weighted kernel mean embedding, using the published bandwidth rule. This analysis uses $\text{SRE-UQ}(\mathbf{r}) = \frac{1}{M} \sum_{i \in \mathcal{T}} |\Delta_i|$, where $\mathcal{T}$ indexes the M highest-amplitude modes. The estimator is imported as written; the contribution is its KG-RAG benchmark, not a new perturbation objective.

VN-Entropy (KG-RAG instantiation). VN-Entropy measures spectral diversity of the normalised response-embedding Gram matrix. With $\mathbf{G} = VV^\top$ and $\rho = \mathbf{G}/N$,

$$\text{VN-Entropy}(\mathbf{r}) = S(\rho) = -\text{tr}(\rho \log \rho) = -\sum_{i=1}^N \lambda_i \log \lambda_i, \quad (7)$$

where $\{\lambda_i\}_{i=1}^N$ are the eigenvalues of ρ . $S(\rho) = 0$ when all responses are identical and approaches $\log N$ when responses are mutually orthogonal. This is a cosine-Gram

instantiation of the VNE family introduced by Kernel Language Entropy (Nikitin et al., 2024); the novelty lies in the KG-RAG benchmark.

SD-UQ (introduced here). SD-UQ is a question-conditioned embedding-dispersion statistic, distinct from Qiu and Miikkulainen (2024)’s Semantic Density. It removes the question direction before measuring residual answer spread: if the sampled answers point in nearly the same residual direction, dispersion is low, the signature of a stabilised (and possibly locked-in) answer surface. Given the unit-norm question embedding $\hat{\mathbf{q}} \in \mathbb{R}^d$, define the orthogonal projector

$$\mathbf{P}_q^\perp = \mathbf{I} - \hat{\mathbf{q}}\hat{\mathbf{q}}^\top. \quad (8)$$

The projected response matrix is

$$\mathbf{V}_\perp = \mathbf{V}\mathbf{P}_q^\perp, \quad (9)$$

where $\mathbf{H} = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ is the row-centering matrix. The thin SVD of the centred projected matrix is

$$\frac{\mathbf{H}\mathbf{V}_\perp}{\sqrt{N}} = \mathbf{U}\Sigma\mathbf{W}^\top. \quad (10)$$

Let $\eta_1 \geq \dots \geq \eta_k$ be the top- k singular values on the diagonal of Σ .

$$\text{SD-UQ}(\mathbf{r}) = \exp\left(\frac{1}{k} \sum_{i=1}^k \log(\eta_i + \varepsilon)\right), \quad (11)$$

where $k = \min(N-1, k_{\max})$ and $\varepsilon = 10^{-12}$ for numerical stability. At $N = 5$, this is a low-sample projected-dispersion statistic, not a full covariance estimate. The geometric mean penalises collapse onto a single residual direction, so a large score requires spread across several retained singular modes. SD-UQ needs only question and answer embeddings; its attraction is operational low cost, not mathematical ornament.

5.2 Retrieval-State Support Diagnostics

Retrieval-state diagnostics ask whether the retrieved graph contains a usable path from the question entities to the asserted answer entity. They matter when $p(c \mid q) \approx \delta_{c^*}$ (Section 3.2): repeated samples then see the same context, so answer-state agreement can be high for both correct and wrong answers. GPS is therefore a graph-support diagnostic reported in risk orientation: lower values mean stronger graph support for the generated answer, while higher values mean weak or absent graph support. It is not a truth label or a logical entailment test.

Let $\mathcal{E}_q$ and $\mathcal{E}_a$ denote the sets of KG entities linked to the question q and answer a , respectively, filtered to $\mathcal{E}_a \setminus \mathcal{E}_q$ to exclude trivial self-loops. Question entities are linked by surface and fuzzy name matching. Answer entities are linked by the same surface matcher *plus* an embedding-based soft matcher: candidate answer spans are embedded with the retrieval encoder and linked to any entity whose name embedding has cosine similarity at least τ to some span. Each linked answer entity carries a link weight w_e : the matched cosine for soft links and $w_e = 1$ for surface matches.

For GPS, answer entity extraction is restricted to the *primary answer span*: the first sentence of the response, truncated to 150 characters, which prevents later explanatory text from inflating the reachable entity count. If no usable answer entity remains after filtering and soft linking, GPS abstains rather than returning a spurious support score.

Graph Path Support (GPS; retrieval-state diagnostic).

GPS asks how strongly the answer is reachable from the question entities within the retrieved KG, and reports the result in risk orientation as one minus that support. Each linked answer entity contributes distance-weighted support $\gamma^{|L_e - \hat{L}(q)|}$, where L_e is the shortest qualifying path length (under the same edge-confidence and provenance filters used at retrieval time), $\hat{L}(q)$ is the expected reasoning depth for the question, and unreachable entities contribute zero:

$$\text{GPS}(q, a) = 1 - \frac{\sum_{e \in \mathcal{E}_a} w_e \gamma^{|L_e - \hat{L}(q)|} \mathbb{I}[e \text{ reachable}]}{\sum_{e \in \mathcal{E}_a} w_e}. \quad (12)$$

GPS = 0 when every linked answer entity is reachable at the expected reasoning depth (full structural support, low uncertainty); GPS = 1 when none are reachable within h hops (no structural support, high uncertainty); with $\gamma = 1$

the score reduces to the unweighted unreachable-entity fraction. The linking threshold and decay were calibrated on the RealMedQA development run ($\tau = 0.60$, $\gamma = 0.4$) and applied frozen elsewhere; a 5×5 sweep over τ and γ (Appendix D.8) shows the held-out AUROCs are stable across the grid, so the reported numbers are not an artefact of the calibrated cell. For 2WikiMultiHopQA and MuSiQue, $\hat{L}(q)$ is the logged per-question hop count; for HotpotQA variants it is the nominal two-hop depth; for RealMedQA it is 1, which makes GPS identical to the calibrated one-hop-decay score on the clinical domain. If the system anchors to the wrong entity but traverses a locally coherent neighbourhood, GPS can still be 0: it scores graph support for the generated answer, not correctness. When no answer entities are found in the primary span (or when the question is a direct-choice comparison), the estimator abstains and the row is excluded from retrieval-state AUROC while being counted in the linking-failure rate (Section 5.4).

5.3 Evidence-State Uncertainty Measures

Evidence-state measures ask whether the retrieved passages support the answer, so they can remain informative when sampled answers agree. They are local answer–evidence consistency scores, not truth labels.

Support Entailment Uncertainty (SEU; evidence-support score). SEU is the normalised entailment–contradiction deficit over retrieved chunks, close to answer-support scoring in RAGs (Es et al., 2024).

Let $c_1, \dots, c_K$ denote the K retrieved chunks and a the generated answer. Each chunk is classified by an NLI model (microsoft/deberta-large-mnli; all model versions are listed in Appendix C): $\ell_k \in \{E, N, C\}$ (entailment, neutral, contradiction) where $E = \text{NLI}(c_k \Rightarrow a)$. Define the support score $s = (n_E - n_C)/K \in [-1, 1]$, where $n_E = |\{k : \ell_k = E\}|$ and $n_C = |\{k : \ell_k = C\}|$.

$$\text{SEU}(q, a, \mathbf{c}) = \frac{1 - s}{2} \in [0, 1]. \quad (13)$$

SEU = 0.0 when all chunks entail the answer (maximum support), SEU = 0.5 when chunks are all neutral (undefined support), SEU = 1.0 when all chunks contradict the answer (maximum conflict). Neutral chunks therefore create a plateau at 0.5, especially when a general-domain NLI model declines to commit on specialist biomedical passages (Section 7.5). Unlike answer-state scores, SEU needs only one generation.

Faithfulness to structured provenance. SEU does not prove that the decoder followed the retrieved graph path. A stricter path-faithfulness diagnostic would align answer claims to entities, relation labels, and relation-anchor text on the retrieved paths. The runs log those ingredients, but not sentence-level claim-to-path alignment for every answer;

path faithfulness is therefore left as a future evidence-state diagnostic.

5.4 Entity-Linking Quality

GPS requires matched question entities and at least one linkable answer entity. Soft answer linking reduces abstention relative to surface matching (Section 7.6); yes/no answers remain a predictable failure case. For backward compatibility the scalar API emits a 0.5 sentinel, but AUROC, ECE, and precision-at- k drop sentinel rows using the recorded null reason.

Entity-linking success rate $g \in [0, 1]$ measures the fraction of content-bearing query tokens covered by matched entity names:

$$g(q) = \min\left(1, \frac{|\hat{\mathcal{T}}_q^{\text{cov}}|}{|\hat{\mathcal{T}}_q|}\right), \quad (14)$$

where $\hat{\mathcal{T}}_q$ is the set of distinct alphanumeric content tokens of length ≥ 4 in q (excluding stopwords), and $\hat{\mathcal{T}}_q^{\text{cov}} \subseteq \hat{\mathcal{T}}_q$ is the subset covered by at least one matched entity name (a token t is covered when it appears as a sub-token of an entity name, or the entity name contains t as a substring). This token-coverage score is a lightweight routing and stratification heuristic, not an uncertainty estimator. Appendix Table 28 summarises the failure regimes of the three families.

6 Experimental Setup

6.1 Datasets

The suite contains six fixed evaluation snapshots over five QA families: biomedical control, clinical QA, and open-domain multi-hop QA (Table 4). It deliberately mixes corpus contracts: per-question bundles, per-question source documents, and shared corpora. The primary graph object is a passage-provenance KG built from the retrieval corpus itself, since the question is whether OntoGraphRAG preserves dense-RAG answer quality while exposing facts, passages, relations, and retrieval state. PubMedQA and MuSiQue use $n = 100$; RealMedQA uses its full $n = 230$ evaluable set; HotpotQA, HotpotQA FullWiki, and 2WikiMultiHopQA use fixed $n = 250$ subsets.

REALMEDQA is the main shared-corpus clinical setting and HOTPOTQA FULLWIKI a controlled Wikipedia shared-corpus stress snapshot; the rest are closed source-document or bundle settings. This is a conservative testbed for KG-RAG: dense retrieval sees small, benchmark-filtered candidate sets and provides a demanding accuracy baseline. PUBMEDQA is retained as a control; the open-domain datasets stress bridge preservation, shared-corpus evidence selection, comparison questions, and deeper compositional chains.

Throughout, hop count refers to the reasoning chain implied by the question, not the number of passages supplied by the benchmark. Explicit decomposition metadata

are used where available, otherwise dataset-specific conventions. Hop-stratified analyses are reported as diagnostics rather than the main table.

Scope. The experiments are fixed-subset diagnostic runs, not leaderboard submissions or web-scale throughput benchmarks. They support the paper’s coarse family-level claims; fine-grained ranking of neighbouring AUROC or AUREC values would require larger multi-seed sweeps.

6.2 Model and Sampling

All answer-generation experiments use GPT-4o-mini (OpenAI, 2024). For answer-state uncertainty, $N = 5$ responses are sampled at temperature $T = 1.0$. These are low-budget black-box diagnostics: the API provides no reliable token likelihoods, so entropy-style scores use sampled responses rather than decoder probabilities.

KG construction is largely deterministic: entity and relation extraction use temperature 0.0, and benchmark passages are ingested one at a time with chunk-level provenance. The studied regime is stabilised entity-first retrieval with explicit fallback.

What is and is not isolated. The comparison holds the corpus, embeddings, and generation model fixed, but does not fully isolate graph structure from retrieval stabilisation, fallback routing, prompt organisation, or provenance grouping (Section 8.1). The runs report mean pairwise chunk overlap across the $N = 5$ calls (Appendix Table 16); an archived 2WikiMultiHopQA trace also supports the stability split in Table 25. The headline runs do not retain per-question entity- or path-overlap statistics for every dataset, but a dedicated graph-state diversity run on HotpotQA-FullWiki logs the full per-sample seed-entity, path, subgraph, and chunk overlap and is analysed in Appendix D.5.

6.3 Correctness Labels

Correctness labels are binary. Label-space tasks are normalised to their answer contract (yes/no/maybe). Free-text and factoid tasks use a reference-based semantic judge given the question, gold answer, aliases where available, and model response; by default the judge is GPT-4o-mini (Appendix C). This is a benchmark answer-normalisation task rather than open-ended clinical diagnosis: the judge decides reference-grounded semantic equivalence, for which exact match would undercount aliases, paraphrases, and short explanatory answers. Same-model judging is a known limitation (Zheng et al., 2023; Panickssery et al., 2024), so the labels are practical reference-grounded judgements rather than an oracle. An independent re-judge with a different model family (Llama-3.3-70B) on the saved answers of every free-text run, including both headline RealMedQA runs, HotpotQA, HotpotQA FullWiki, MuSiQue, and both 2Wiki-

Table 4: Evaluation datasets, corpus contracts, KG scale, and primary roles. h is the KG traversal depth; *corpus* denotes the retrieval contract; n is the evaluated subset size. $|E|$ and $|R|$ count entity nodes and typed entity–entity relations in the dataset-scoped KG, queried from the persistent Neo4j stores used in the reported runs.

Dataset	Domain	Answer type	h	Corpus	n	$ E $	$ R $	Primary role
PubMedQA	Biomedical	Yes/No/Maybe	2	abstract	100	2,354	3,023	Binary control; strong dense baseline
RealMedQA	Clinical	Free-text	2	shared (143)	230	537	359	Clinical shared- corpus grounding
HotpotQA (Yang et al., 2018)	Wikipedia	Free-text	2	bundle	250	13,546	12,355	Open-domain bridge diagnostic
HotpotQA FullWiki (Yang et al., 2018)	Wikipedia	Free-text	2	shared subset	250	13,498	12,012	Shared-corpus bridge stress test
2WikiMultiHopQA (Ho et al., 2020)	Wikipedia	Free-text	2	bundle	250	8,328	7,187	Bridge and compari- son analysis
MuSiQue (Trivedi et al., 2022)	Wikipedia	Free-text	4	bundle	100	15,440	24,858	Hard multi-hop stress test

MultiHopQA runs, agrees with the original labels on 92–99% of answers ($\kappa = 0.52$ – 0.97 , the low end being the near-ceiling-accuracy RealMedQA adaptive run where few wrong answers make κ unstable) and leaves every central contrast unchanged (Section 8.1).

Two conventions matter. Provider-side failures are recorded explicitly, so results report both **raw accuracy** and **clean accuracy**, the latter excluding generation failures (Appendix Table 13). For free-text datasets, prompts elicit short explanatory answers rather than extractive spans; semantic correctness is therefore the headline answer metric, with EM/F1 retained only in the artefacts.

6.4 Evaluation Metrics

The main uncertainty metric is **AUROC**, the ranking quality for separating correct from incorrect answers. **AUREC** is logged as a secondary selective-prediction audit metric (Geifman and El-Yaniv, 2017), but not expanded into a parallel results table. Raw accuracy, clean accuracy, retrieval overlap, and per-metric compute time are also logged; the marginal cost of each diagnostic, including SEU’s per-chunk NLI calls and GPS, is reported in Appendix Table 20. All main figures report point estimates from one fixed subset per dataset. With mixed sample sizes ($n = 100$, $n = 230$, $n = 250$), small AUROC gaps are descriptive unless they recur across signal families or match the qualitative traces. No multiple-comparison correction is applied because the heatmap is not used for family-wise significance testing. The two primary headline contrasts instead carry paired-bootstrap intervals; the largest, the adaptive-versus-strict SD-UQ collapse of +0.52 AUROC at $n = 196$, is far from zero, whereas gaps near 0.05 are not interpreted. Appendix Tables 12, 14, 21, and 24 document run configurations, answered counts, GPS usable counts, the HotpotQA FullWiki snapshot, and bootstrap intervals. Retrieval-state AUROC with fewer than roughly fifty usable rows is treated as trace-level evidence. GPS AUROC is always conditional

on rows for which GPS is defined; linking failures and other abstentions are reported through usable/answered denominators and audit-rule coverage, not hidden inside the AUROC.

The headline tables report family representatives: DSE, SRE-UQ, VN-Entropy, and SD-UQ for answer state; SEU for evidence state; GPS for retrieval state. The P(True)-style proxy is omitted because it is monotone in DSE here, and SelfCheckGPT is retained in the artefacts. GPS uses surface plus embedding-based soft answer-entity linking; its two hyperparameters were calibrated on the RealMedQA development run and frozen elsewhere. All GPS values were recomputed by post-hoc replay on saved answer logs without rerunning generation, retrieval, or entity linking (Appendix C). PubMedQA-style yes/no answers remain outside GPS’s intended scope because they expose no linkable answer entities.

6.5 Retrieval and KG Configuration

Both systems use all-MiniLM-L6-v2. Vanilla RAG performs direct vector retrieval over chunks. KG-RAG performs entity-first retrieval, graph expansion, and context assembly from graph-linked chunks, paths, and entities, with dense retrieval and retriever-first graph expansion as fallbacks when entity anchoring is weak. Each KG-RAG response records its route (`entity_first`, `rfge`, or `semantic_only`) and route reason.

Dataset-scoped KGs are built passage-wise, preserving question and passage provenance and avoiding artificial cross-question chunking. Each artefact retains chunk identifiers, passage titles where available, relation paths, route labels, and relation-anchor text when supplied by the KG extractor. Dense retrieval can log passages and scores; the KG layer additionally exposes matched entities, typed triples, graph paths, relation anchors, and abstention events. The traceability claim is therefore structured provenance, not provenance in the abstract. Traversal depth is dataset-specific: $h = 2$ for PubMedQA, RealMedQA, HotpotQA,

HotpotQA FullWiki, and 2Wiki; $h = 4$ for MuSiQue. During iterative decomposition, each sub-question uses a bounded local graph expansion cap so that overall chain length and per-step search breadth are not confounded.

7 Results

The results are organised around the lock-in mechanism, not leaderboard position. They start with accuracy, then follow the answer-, evidence-, and retrieval-state signals through adaptive retrieval, the strict graph-only stress test, and the composite audit. Table 5 gives the running summary; intervals and denominators are kept in the appendix. The recurring pattern is simple: answer-state metrics rank errors well in ordinary adaptive runs, but can assign near-zero uncertainty to repeated wrong answers under lock-in (the failure taxonomy in Table 6). GPS is reported more cautiously, as a conditional, domain-dependent diagnostic with its usable denominator shown throughout.

7.1 Accuracy: Dense Retrieval Leads; Parity Is the Bar to Clear

Table 5 makes the accuracy picture plain: KG-RAG trails dense retrieval on five of the six snapshots, with Δ from -0.004 (RealMedQA, effective parity) to -0.084 (MuSiQue), so it neither beats nor reliably matches dense retrieval on answer quality, and OntoGraphRAG is no drop-in accuracy replacement on the harder multi-hop suites. Where parity holds ($|\Delta| \leq 0.023$ on PubMedQA, RealMedQA, and 2WikiMultiHopQA), the diagnostic question becomes sharp: what does the graph expose beyond passages? The adaptive runs test whether KG-RAG preserves accuracy while exposing anchors, paths, and provenance; the strict graph-only stress test forces retrieval concentration to probe lock-in.

7.2 Silent Failures Under the Deployable Policies

Before comparing estimators, the deployable policies already show why answer-state uncertainty cannot be the whole story. Table 7 reports the *silent-failure rate*: the fraction of wrong answers whose answer-state uncertainty is exactly zero ($DSE = 0$ and $SD-UQ$ at its numerical floor), so that no score using only within-question answer dispersion has a disagreement signal to exploit. Requiring both conditions is deliberately conservative. $DSE = 0$ is the cluster-count view, where all samples fall in one semantic cluster. The $SD-UQ$ floor is the embedding-geometry view, where the geometric-mean dispersion collapses onto the $\varepsilon = 10^{-12}$ stabiliser of Equation (11). A row counts as silent only when both views register no residual variation. The blind spot belongs to the whole family, not to one estimator: identical samples contain no disagreement signal for DSE , $SD-UQ$,

VN -Entropy, or any answer-dispersion variant, though signals outside that family may still help. Under the deployable policies, 42% of adaptive KG errors and 59% of dense-retrieval errors are silent, rising to 84% under the strict ablation. Because the definition requires both semantic unanimity and the $SD-UQ$ numerical floor at $N = 5$, these fractions are conservative symptom counts for this answer-state family, not exact estimates of every undiagnosable error.

Confidently wrong answers are common in these deployable-policy runs, not only in the stress regime. Dense retrieval even produces *more* silent errors than adaptive KG-RAG, so the phenomenon is not specific to graph retrieval; nor is the dense silence a low-budget artefact, since on dense 2WikiMultiHopQA the silent rate is still 68% at $N=20$. The pattern likewise persists under relaxed silent-error floors (Appendix D.1).

The KG-specific risk is *auditable wrongness*: a silent error can arrive with matched entities, typed paths, and provenance anchors, making the system look structurally auditable while it is wrong. Dense retrieval exposes passages, but not that symbolic trace. As established in Section 1, silence is a symptom count; route logs are needed to diagnose lock-in.

In plain terms, an answer-dispersion method cannot recover errors that never disagree. At the operational $N=5$ budget, a method that only observes sampled-answer disagreement can recall at most 41% of dense errors, 58% of adaptive KG errors, and 16% of strict-stress-test errors; the remaining wrong answers provide no within-question disagreement signal to rank. That missing observable motivates evidence-state and retrieval-state diagnostics.

7.3 Answer-State Metrics Are Strong, but Not Sufficient

Given this blind spot, the next question is how much answer-state estimators still contribute when retrieval remains adaptive. In the adaptive runs they remain the strongest default: the AUROC heatmap (Figure 3) shows that answer-side scores have the clearest overall association with error, especially when retrieval and decoding still produce meaningful answer variation. Their limit is equally clear. When graph routing stabilises the context around a wrong entity or path, answer disagreement can be low for incorrect answers; the score then measures the smoothness of the answer surface rather than the reliability of the retrieved state. Answer-state uncertainty is therefore a strong default, but not a lock-in diagnostic.

Within that family, the embedding-geometric scores introduced here are the best practical answer-side baselines in these low-sample runs. VN -Entropy and $SD-UQ$ match or beat the imported semantic-entropy and perturbation baselines on five of the six KG snapshots, with the exception confined to MuSiQue; the per-dataset AUROCs and intervals are in Appendix Table 24. The stability split explains why: under high retrieval overlap, DSE falls to chance while

Table 5: Headline accuracy and KG-side diagnostic AUROC. Vanilla RAG uses dense retrieval; KG-RAG uses entity-first graph retrieval with fallback. Accuracy is clean semantic accuracy after provider failures are excluded; n counts answered KG rows and Δ is KG – dense accuracy. SRE/SD are answer-state scores, SEU is evidence-state, and GPS is the soft-linked, depth-matched graph-support diagnostic, reported in risk orientation (lower is stronger support), shown with its usable/answered denominator. Intervals and denominator details are in Appendix C.

Dataset	n	Dense acc.	KG acc.	Δ	SRE	SD	SEU	GPS (usable)	Main reading
PubMedQA	100	0.750	0.730	−0.020	0.64	0.70	0.59	–	Binary control; GPS abstains (no answer entities).
RealMedQA	223	0.950	0.946	−0.004	0.66	0.81	0.50	0.76 (195/223)	Clinical parity; GPS calibration domain.
HotpotQA	238	0.655	0.605	−0.050	0.63	0.64	0.48	0.51 (158/238)	Dense stronger; KG adds anchors and paths.
HotpotQA FullWiki	218	0.721	0.665	−0.056	0.69	0.67	0.56	0.54 (185/218)	Shared-corpus stress; GPS near chance.
2WikiMHQA	244	0.712	0.689	−0.023	0.65	0.69	0.64	0.38 (182/244)	Near parity; GPS weak on bridges.
MuSiQue	94	0.478	0.394	−0.084	0.66	0.63	0.63	0.68 (83/94)	Hard multi-hop; auditable, not competitive.

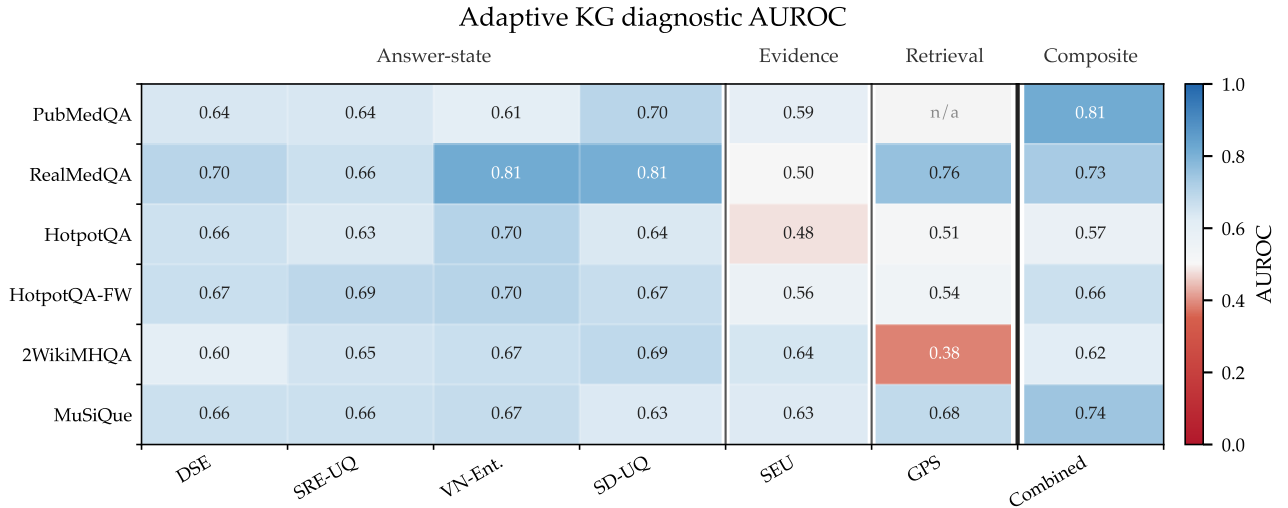


Figure 3: Adaptive KG-side diagnostic AUROC across the six snapshots, grouped by family. Incorrect answers are the positive (risk) class. Vertical rules separate the answer-state, evidence-state, and retrieval-state families; the derived Combined column is separated because it is not a raw metric, but the mean of within-dataset percentile ranks for SD-UQ, SEU, and GPS-risk. GPS denominators are in Appendix Table 14; PubMedQA is undefined because yes/no answers expose no linkable entities. The diverging colour scale is neutral at chance (0.50), red below, blue above. Cells are point estimates; bootstrap 95% CIs are in Appendix Table 24.

SD-UQ retains signal by removing the question direction before measuring residual dispersion (Appendix Table 25). SD-UQ is a useful black-box baseline: cheap, embedding-only, and available without log-probabilities, NLI calls, or model internals.

The verbalised-confidence baseline confirms the boundary rather than removing it. Because it asks the model for a probability, it is not blind to silent errors by construction and recovers part of the clinical silent-error mass. It still falls to chance on bridge-style lock-in and returns a number without auditable provenance; the full comparison is in Appendix D.6.

7.4 The Strict Stress-Test Collapse

Answer-state metrics hold up under adaptive retrieval; the question is whether they survive when retrieval is deliberately concentrated. The sharpest stress-test signature appears in the strict graph-only regime, where the retrieved graph state is made more concentrated by disabling dense fallback, augmentation, reranking, and decomposition, and answer-state uncertainty becomes a poor guide to correctness. That answer dispersion falls under forced concentration is expected. The informative part is what the collapse is made of, which the route decomposition below recovers.

Across all paired adaptive-strict questions, 43/196 RealMedQA questions and 87/242 2WikiMultiHopQA questions move from correct under adaptive KG to wrong-and-silent under the strict graph-only stress test. Among

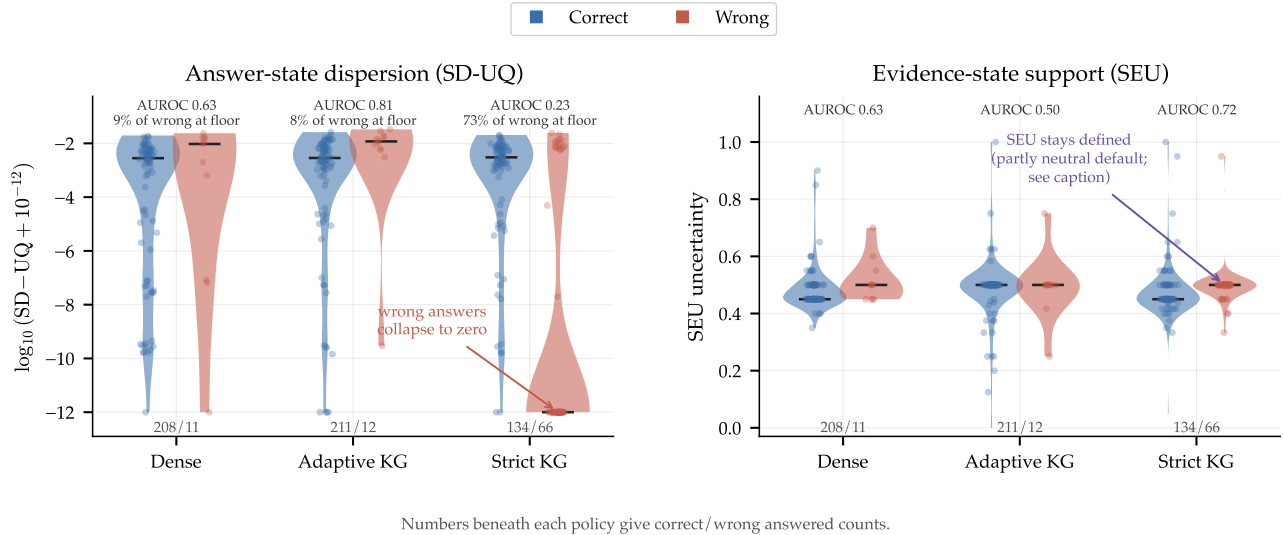


Figure 4: Context collapse in the RealMedQA strict graph-only stress test. Violin/strip distributions are split by correctness, with AUROC printed in each panel. Under strict retrieval, 73% of wrong answers collapse to the SD-UQ floor (48/66 vs. 1/12 adaptive; Fisher exact $p = 4 \times 10^{-5}$). SEU remains separated, although Table 8 shows that much of the strict signal comes from empty-retrieval defaults. The counts beneath each policy are correct/wrong answered rows. Strict graph-only retrieval is a mechanism probe.

questions already wrong but not silent under adaptive KG, another 2 RealMedQA and 12 2WikiMultiHopQA questions become wrong-and-silent under strict retrieval; on 2WikiMultiHopQA, 31 questions remain wrong-and-silent under both policies. Thus the stress test does not merely lower accuracy: it moves errors into the part of the state space where answer-dispersion methods have no disagreement signal to exploit.

A strict RealMedQA stress test makes the lock-in regime visible directly on the cleanest parity dataset. As scoped in Section 1, it is a regime probe: it shows that lock-in can arise and what the silent signature looks like. The adaptive rates in Section 7.2 measure the deployable-policy footprint. It is deliberately harsher: clean KG accuracy falls to 0.670 on 200 answered questions, and average within-question chunk overlap rises from 0.558 to 0.661 (Appendix Table 16). The danger is stability around a wrong symbolic state, not retrieval stability in itself. Under that concentrated regime, answer-state ranking largely collapses, as the strict/adaptive comparison in Figure 4 shows: DSE AUROC is 0.463, VN-Entropy AUROC is 0.248, and SD-UQ AUROC is 0.233. On the question subset shared by the two policies, the adaptive-to-strict SD-UQ gap is +0.52 AUROC (paired-bootstrap 95% CI [0.28, 0.73], $n = 196$). The migration plot in Figure 5 shows the mechanism at question level: wrong answers migrate from moderate output dispersion under the adaptive policy into a dense column at the SD-UQ floor under strict retrieval, while their evidence-support scores remain spread. That matters because the low-dispersion corner is exactly where answer-state confidence looks most reassuring and is least informative. By contrast, SEU reaches 0.721 AUROC, and GPS reaches 0.746 on

142/200 non-abstained rows (Appendix Table 23): the answer surface goes silent while retrieval-facing scores still rank errors (GPS here is in its calibration domain). The contrast holds under an independent Llama-3.3-70B judge (95.0% agreement, $\kappa = 0.89$): SD-UQ stays collapsed while SEU and GPS still rank the errors, which same-model self-preference would not explain (Section 8.1).

Route decomposition. First, the collapse is not a chunk-level artefact. An interventional dose-response reduces chunk overlap from 0.67 to 0.29, but leaves SD-UQ AUROC flat (0.18) and the silent-error count unchanged (Appendix D.2).

Second, route logging shows where the silent state lives. Every silent error in the strict clinical run is an empty-retrieval row (48/48 at $n = 230$). Strict anchoring found no usable context, the same empty state recurred across all five samples, and the generator answered unanimously from parametric memory. This points first to removed fallback and vector augmentation as proximate contributors in this stress run, but the design still changes multiple components and does not isolate one as the cause. On the populated-route rows, answer-state ranking is healthy (SD-UQ AUROC 0.79). Much of SEU’s apparent survival is mechanical: empty-retrieval wrong rows receive the all-neutral default 0.5, which ranks above confidently supported correct rows; populated-row SEU is only 0.56. This is *absence* lock-in, whose simplest reliable flag is the empty-route/abstention observable, not answer dispersion or evidence entailment.

Third, the strict 2WikiMultiHopQA run decomposes the same way but leaves a remainder: 110 of 130 silent errors are empty-retrieval, while 20 ride populated entity-first

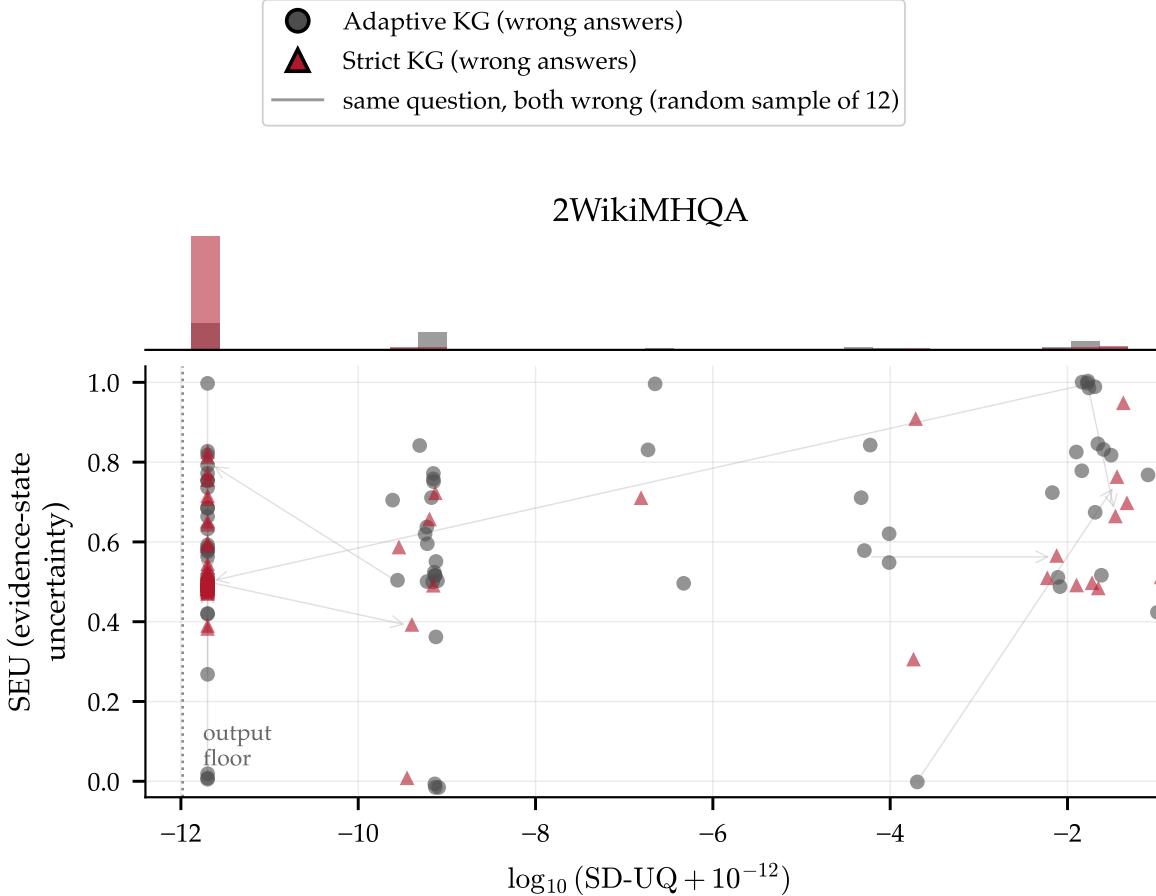


Figure 5: Lock-in as migration into the low-dispersion corner (2WikiMultiHopQA). Each point is a *wrong* KG answer, plotted by answer-state dispersion (SD-UQ, log scale) and evidence-state uncertainty (SEU), under the adaptive policy (grey circles) and the strict graph-only stress test (red triangles). Grey arrows connect a uniformly random sample of 12 questions that are wrong under *both* policies. Under strict retrieval the SD-UQ mass collapses onto the floor while SEU remains spread. Small vertical jitter is added for legibility.

routes, wrong-but-coherent neighbourhoods of the Osireion type (*presence* lock-in), and populated-row SD-UQ stays degraded there (0.65). Presence lock-in is the harder case: the answer surface and the route observable are both silent, leaving only evidence-level checks. It is also a setting where SEU itself proved unreliable (the strict 2-hop slice), which is why the composite recommendation does not reduce to any single surviving family.

Under the deployable adaptive policy the silent errors are mostly presence-like: of the 141 pooled adaptive silent errors, 99 carry recorded route metadata, and among those 89 ride populated iterative multi-hop routes with none on the empty-retrieval route, so the deployable-policy footprint is not an empty-retrieval artefact. The route claim is limited to the logged subset: the remaining 42 silent errors lack route metadata, so their absence/presence mix is not recoverable from the saved logs. Decomposing those 141 adaptive silent errors by which family flags them: 67 carry an evidence-state contradiction (SEU > 0.5), 127 carry weak or absent graph support (GPS > 0.5 or abstention), and only 7 (5%)

remain unflagged by every non-answer family. A manual audit of these seven (Appendix D.9) finds mostly benchmark-ambiguous questions, wrong-answer-slot errors, and parametric confluents rather than confirmable deep presence lock-in whose evidence *entails* the wrong answer (the Baltic Cup type of Table 30); since retrieved chunk texts are not archived, that deceptive type cannot be confirmed from logs.

Flagging the cases where all three families fail, with SEU and GPS both near zero on a wrong answer, would need exactly what the logs lack here: archived chunk text for cross-passage contradiction mining, and relation-level path faithfulness to check that the connecting path uses the relation the question asked for rather than a locally coherent but wrong one. Both are concrete, retrieval-replay-only extensions, not new theory. Table 8 makes the accounting explicit.

Under this deterministic graph-only regime, answer-surface uncertainty can cease to track correctness. Which signal survives depends on the variant: route or abstention observables for absence, and evidence-level checks, imperfectly, for presence.

Table 6: Failure taxonomy for retrieval-augmented generation. Retrieval-state lock-in (top right) is the focal regime: the retrieval state is defective while answer-state uncertainty is low. Absence lock-in is flagged by retrieval-state abstention; presence lock-in, a coherent wrong neighbourhood, remains the hard case. At $N=5$, 42% of adaptive-KG errors fall in this silent cell (59% dense, 84% strict).

	Retrieval state correct	Retrieval state wrong / missing
Low answer-state uncertainty	Certified low-risk all three families concur	Retrieval-state lock-in answer-state silent; absence variant flagged by retrieval-state abstention, presence variant open (<i>this paper</i>)
High answer-state uncertainty	Ordinary uncertainty answer-state flags it	Retrieval failure answer-state flags it; retrieval-state abstains or shows no support

Table 7: Silent-failure rates with Wilson 95% intervals. Fraction of wrong answers with zero answer-state uncertainty (DSE = 0 and SD-UQ at its numerical floor), per retrieval policy. n is the number of wrong answered rows.

Dataset	Dense	Adaptive KG	Strict KG
PubMedQA	.84 [.65,.94] (25)	.67 [.48,.81] (27)	–
RealMedQA	.09 [.02,.38] (11)	.08 [.01,.35] (12)	.73 [.61,.82] (66)
HotpotQA	.59 [.48,.69] (78)	.46 [.36,.56] (94)	–
HotpotQA-FW	.45 [.33,.58] (58)	.18 [.11,.28] (73)	–
2WikiMHQA	.79 [.67,.87] (61)	.55 [.44,.66] (76)	.90 [.84,.94] (145)
MuSiQue	.51 [.37,.65] (47)	.42 [.30,.55] (57)	–
Pooled	.59 [.53,.65] (280)	.42 [.36,.47] (339)	.84 [.79,.89] (211)

The same pattern appears on the newer 2WikiMultiHopQA $n = 250$ run when the three retrieval policies are compared on the identical question subset; the full hop-stratified breakdown is in Appendix Table 29. The adaptive KG policy is the appropriate deployment comparison: it remains close to dense retrieval overall (clean accuracy 0.689 versus 0.712) and matches dense retrieval on the 4-hop slice. On the 4-hop slice, strict graph-only KG has zero clean accuracy while SD-UQ and VN-Entropy both collapse essentially to zero: the answer surface is stable precisely when the retrieved graph state has become wrong enough to make every answer fail. Low answer-state uncertainty is not intrinsically bad. In the dense 4-hop row, SD-UQ is also near zero, but VN-Entropy remains non-zero and clean accuracy stays high, so the collapse signature is the joint collapse of SD-UQ and VN-Entropy together with failed answers, not low SD-

UQ alone. For KG-RAG, answer agreement must therefore be read alongside evidence- and retrieval-state diagnostics when the symbolic state has concentrated around a wrong neighbourhood. Per-question support comes from a dedicated graph-state diversity run on HotpotQA-FullWiki that logged per-sample overlap for the seed entities, paths, subgraph, and chunks ($n = 221$ answered, 74 wrong; Table 9). Across the $N = 5$ samples the retrieval state is highly stable (seed-entity Jaccard 0.99; path, subgraph, and chunk 0.58–0.65), so repeated samples really do condition on nearly the same state; and among *wrong* answers more overlap predicts lower dispersion (path/subgraph/chunk Spearman $\rho = -0.30/-0.23/-0.24$, all $p < 0.05$, $n = 74$; the seed-entity coupling is null only because seed overlap saturates near 1.0, leaving no variance to correlate against). The archived traces show the same sign, carried by the bridge-style 2WikiMultiHopQA trace ($\rho = -0.32$, $p = 0.01$, $n = 60$; pooled $\rho = -0.27$), with no comparable pattern on the dense side. These analyses support the mechanism without identifying it causally (Appendix D.5, D.3; the per-dataset concentration facets are in Figure 7).

HotpotQA supplies a useful counterweight. Dense retrieval leads on accuracy (0.655 vs. 0.605; FullWiki 0.721 vs. 0.665) while answer-state AUROCs stay healthy on both systems and GPS is only weakly discriminative (0.51 and 0.54 usable-row AUROC; Appendix Table 21). These runs mark the thesis boundary rather than contradicting it: when dense retrieval already has a compact, well-indexed candidate set, graph organisation adds provenance and traceability but the current policy does not improve answer quality or retrieval-state ranking.

Figure 6 reframes the same results as a selective-prediction problem, the clinical form of the uncertainty question: if a diagnostic score is high, should the system answer, broaden retrieval, or send the case to review? In ordinary adaptive runs, answer-side scores often give the cleanest rejection signal. In lock-in regimes, SEU and GPS have to be read as provenance checks rather than weaker versions of answer-state entropy.

The key appendix row is the strict graph-only 4-hop slice ($n = 58$, clean accuracy 0.000, SD-UQ mean 0.000, VN-Entropy mean 0.000). AUROC is intentionally left undefined because all 58 clean answers are incorrect, so the evidence is the conjunction of zero accuracy and collapsed output scores, not a forced ROC statistic: a mechanism demonstration, not a deployment result.

7.5 Evidence-State Metrics Add Support-Level Signal

If the answer surface goes silent under concentration, the next question is whether the retrieved evidence itself betrays the error. Evidence-state measures ask whether the retrieved passages support the answer that was produced, not whether sampled answers agree; this matters especially in high-accuracy or near-parity settings where the deployment

Table 8: Silent-error accounting from saved logs. Buckets are mutually exclusive and apply only to wrong answered rows. “Max answer recall” is the structural upper bound for any answer-dispersion-only detector at the $N=5$ budget: silent wrong answers contain no sampled-answer disagreement signal. Empty denotes absence lock-in; Pop.+SEU and Pop.+GPS are populated-route silent errors flagged by evidence or graph-support signals. Unflagged is the residue with no non-answer-family flag.

Run	Wrong	Silent	Max answer recall	Empty	Pop.+SEU	Pop.+GPS	Route-unk.+flag	Unflagged
Adaptive KG pooled	339	141	.58	0	40	44	50	7
Strict KG pooled	211	178	.16	158	13	6	0	1
RealMedQA strict	66	48	.27	48	0	0	0	0
2WikiMHQA strict	145	130	.10	110	13	6	0	1

Table 9: Per-sample graph-state diversity on the dedicated HotpotQA-FullWiki run ($n = 221$ answered, 74 wrong). *Stability* is the mean (median) pairwise Jaccard overlap across the $N = 5$ samples per question, by retrieval-state family. *Coupling* is the correlation between that per-question overlap and answer-state dispersion (SD-UQ) among wrong answers ($n = 74$), reported as Spearman’s ρ with its two-sided p -value; a negative value means more overlap goes with less dispersion, the lock-in prediction.

Family	Stability (Jaccard)		Coupling vs SD-UQ	
	Mean	Median	ρ	p
Seed entity	0.99	1.00	+0.09	0.47
Path	0.58	0.58	−0.30	0.009
Subgraph	0.64	0.63	−0.23	0.048
Chunk	0.65	0.65	−0.24	0.037

question is whether the answer can be justified by retrieved evidence.

The aggregate evidence-state signal is weaker than the answer-state family as a global ranker, but complementary rather than redundant. SEU’s survival under lock-in is dataset-dependent. It reaches 0.721 AUROC on the strict RealMedQA ablation, although the route decomposition in Section 7.4 shows that much of this signal comes from the all-neutral default over empty-retrieval rows, not from genuine entailment. On the strict 2WikiMultiHopQA 2-hop slice, it falls below chance (Appendix Table 29). Evidence support is therefore useful, but not uniformly robust to retrieval concentration. The 2-hop failure has a concrete mechanism: on bridge questions the strict policy retrieves passages about the wrong-but-coherent anchor, and those passages weakly *entail* the wrong answer drawn from that neighbourhood rather than contradicting it, so low SEU and incorrectness coincide. On the clinical corpus, the contradiction is visible in the retrieved text itself; SEU survives there for this reason. SEU is also low-resolution where retrieval returns long, loosely-relevant chunks: on the adaptive RealMedQA run, 74% of answered rows receive the all-neutral value 0.5 because no retrieved chunk is judged to entail or contradict the answer, which explains the chance-level adaptive AUROC there. In that regime the score is silent rather than wrong; sentence-level entailment scoring is the plausible refinement, but it requires re-running retrieval rather than replaying logs. The selective-prediction curves in Figure 6 and the fixed-coverage operating points in Appendix Table 17 give the same lesson in decision form.

Under the adaptive policy, gating on SD-UQ alone is the strongest simple abstention rule; under the strict clinical ablation that rule becomes uninformative (0.338 error at 80% coverage against a 0.330 no-abstention base) while SEU gating cuts the residual error to 0.269, an 18% relative reduction in the regime where the answer surface is silent. The useful diagnostic changes with the retrieval regime; a single global ranking hides that switch.

7.6 Where Retrieval-State Diagnostics Work and Where They Do Not

Retrieval-state diagnostics ask the complementary question: when does the graph state itself reveal weak support? GPS behaviour is domain-dependent. On the clinical shared-corpus setting where its two hyperparameters were calibrated, it is a useful error ranker (0.76 adaptive, 0.75 strict); on the held-out open-domain suites transfer is uneven (MuSiQue 0.68 with per-question depth, but HotpotQA 0.51, FullWiki 0.54, and 2WikiMultiHopQA 0.38 near or below chance). A sensitivity replay over $\tau \in \{0.50, \dots, 0.70\}$ and $\gamma \in \{0.2, \dots, 1.0\}$ leaves the held-out open-domain GPS AUROCs essentially unchanged ($SD \leq 0.018$ for HotpotQA, FullWiki, 2WikiMultiHopQA, and MuSiQue; Appendix Table 27), so the weak 2WikiMultiHopQA ranking is not a threshold/decay artefact. GPS scores graph support for the generated answer, not correctness. It can therefore rate a wrong-but-coherent neighbourhood as low risk: in the Baltic Cup case of Table 30, the wrong entity is reachable, GPS is 0.00, SEU is 0.00, and the answer surface has near-zero dispersion. The decomposition makes that failure visible but does not solve it. Retrieval-state AUROC is therefore computed only on rows with linkable answer entities, with linking failures reported as abstentions (Table 5). GPS contributes most reliably as an abstention record, usable denominator, and trace-level support check; its scalar ranking is useful only in validated domains. A fixed-bin reliability check makes the same point without AUROC: RealMedQA error rates rise from 0/39 in the lowest GPS-risk bin to 6/42 in the highest, but the open-domain bins are not monotone, especially on 2WikiMultiHopQA (Appendix Table 15).

Soft answer-entity linking substantially reduces the abstention cost: usable denominators now cover most answered rows (e.g. 185/218 on HotpotQA FullWiki, up from

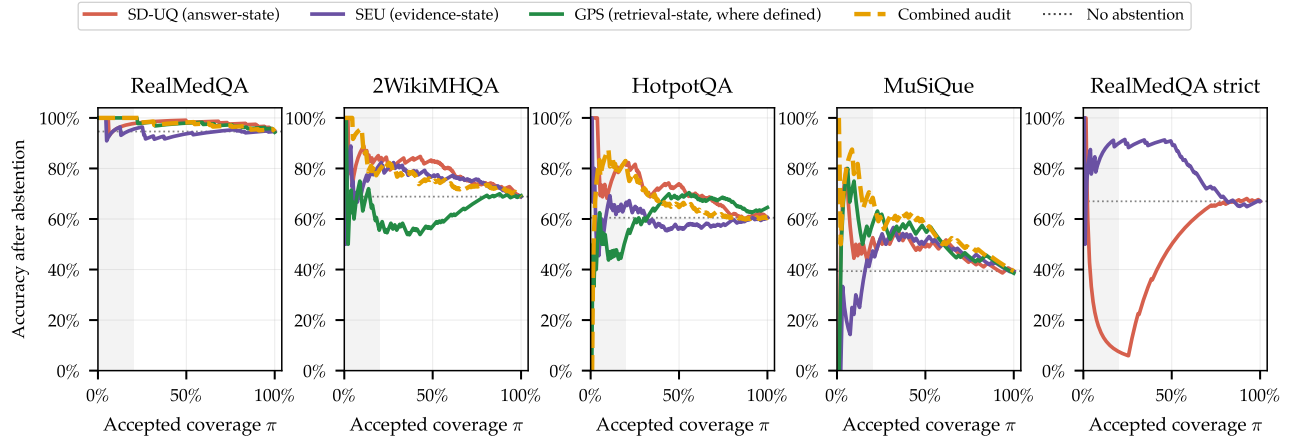


Figure 6: Selective prediction with different KG-side diagnostic families, including the RealMedQA strict ablation (rightmost panel). Curves sort answered KG-RAG queries by increasing uncertainty and then report the accuracy retained at each accepted coverage level; the dotted line is the no-abstention accuracy and the shaded band marks the noisy $< 20\%$ -coverage region. The strict panel carries the decision-form headline: SD-UQ gating *inverts*: accepting the lowest-dispersion answers first means accepting the silent wrong mass, while SEU gating climbs toward 90%. GPS and the combined audit are omitted from the strict panel because GPS comes from post-hoc replay rather than row-level logs.

69 under pure surface matching), with the per-dataset values in Table 5 and Appendix Table 14. PubMedQA still abstains completely because binary answers expose no answer entities, a structural boundary rather than a linking failure. Coverage repair is not ranking repair: the held-out AU-ROCs above show that making more rows scoreable does not by itself make the score a better error ranker. Closing the open-domain ranking gap most plausibly requires relation-level path faithfulness, checking that the connecting path uses the relation the question asked for, rather than further entity-linking improvements. The 2WikiMultiHopQA replay shows the problem directly. The adaptive $n = 250$ run gives GPS AUROC 0.38 on 182/244 usable rows because 71% of correct adaptive answers have their gold entity unreachable in the retrieved subgraph, against 45% of wrong ones. Here GPS is not miscalibrated; it is exposing a retrieval policy that can make the wrong answer reachable while leaving the gold answer outside the graph state.

The post-hoc replay on the full $n = 230$ answer log, re-computing only GPS, without rerunning generation or retrieval, raises KG-side coverage from 97/223 to 195/223 and AUROC from 0.47 to 0.76 (Appendix Table 22); the same replay reaches 0.89 on the dense side. RealMedQA is the calibrated case: it shows what retrieval-state scoring can deliver on a clinical shared corpus, while the held-out rows above show the generalisation limit.

7.7 Concrete Cases

The aggregate AUROCs and route statistics summarise behaviour across questions; the Osireion trace in Figure 1 shows how the three families interact on a single answer, and is deliberately not a case where every diagnostic suc-

ceeds. Answer-state uncertainty is at its floor ($DSE = 0$, $SD-UQ \approx 0$) because all samples repeat the wrong entity; GPS is low (0.44), so it does not flag the case: the wrong entity is reachable in the retrieved graph. This is correct behaviour for a risk-oriented support diagnostic, but not a truth guarantee. SEU alone raises risk (1.0), because every retrieved chunk contradicts the confident, graph-coherent answer. The practical meaning of path faithfulness is therefore that the graph can make a wrong reasoning chain visible even when the scalar retrieval-state score reports low risk, provided the evidence-support head is read with it; the full per-case audit readout is in Appendix Table 31. A consolidated case gallery, with seven verified lock-in failures, four KG successes, and per-case diagnostic readouts, is in Appendix E (Table 30).

7.8 The Composite-Suite Thesis

Finally, if the three families answer different diagnostic questions, the decision layer should not collapse them too early. They also barely move together, which is the quantitative case for keeping them apart; Appendix Figure 8 shows this for the adaptive KG setting. Across the six-snapshot heatmap suite ($n = 803$ pooled non-abstained KG rows), the pooled KG-side Spearman correlations are small to modest: $\rho = 0.03$ for SD-UQ versus SEU, $\rho = -0.27$ for SD-UQ versus GPS, and $\rho = -0.10$ for SEU versus GPS; the pooled family-disagreement scatter is in Appendix Figure 8. The three scores therefore carry partly separate diagnostic signal, even if low correlation alone cannot identify distinct latent causes. Because GPS is a graph-support diagnostic reported in risk orientation rather than an information-theoretic uncertainty score, lower GPS means stronger local graph sup-

port for the generated answer, so the negative SD-UQ–GPS sign should not be read as a universal risk law.

If all three families tracked the same object, a single scalar confidence score would suffice; they do not, and expose different failure modes. The recommendation is a composite diagnostic suite, not a single-metric leaderboard. In practice, a low answer-state score should not override a retrieval-state abstention, a fragile graph path, or high evidence contradiction; such discordant profiles are good candidates for abstention, broader retrieval, or human review. A deliberately simple percentile-rank audit score, averaging SD-UQ, SEU, and GPS-risk within each dataset, shows how the three families can feed a decision layer. It is uneven, but useful where answer dispersion is coarse (e.g. PubMedQA Combined 0.813 vs. SD-UQ 0.699; Appendix D.7, Table 26).

A conjunctive low-risk audit rule. The conservative rule is narrow by design: it selects 7.7% of answers at 91.9% accuracy (86 of 1,117 questions), against a 69.7% base rate. It treats a response as low-risk only when all three families concur: low answer-state uncertainty (SD-UQ in the lower half of its dataset distribution), non-contradictory evidence support ($\text{SEU} \leq 0.5$), and defined, present retrieval-state support ($\text{GPS} \leq 0.5$). The cutoffs are label-free: SEU and GPS use their fixed neutral point (0.5), and SD-UQ uses the dataset median. A deployment would choose its own coverage point on held-out calibration data. On the clinical target domain it selects 22% of answers at 100% precision under the automated judge (48/48 correct), a figure that would need human-expert confirmation before any clinical use (Section 8.2). The same fixed policy is applied unchanged to every dataset. Table 10 therefore acts as the cross-dataset generalisation view: it reports per-dataset coverage, precision with Wilson intervals, and recall. Recall is included because a high-precision selection rule is trivial without it. Coverage is structurally limited: the rule cannot select answers with no linkable answer entity (binary, numeric, or label-space answers, where GPS abstains by construction), so on those formats it defers rather than selects, and a deployment would need a separate fallback. The conjunction beats matched-coverage scalar gates: SD-UQ alone gives 81.4% precision at the same coverage, and the percentile-rank combined score gives 88.4%. A dataset-stratified Mantel–Haenszel check gives the same direction (odds ratio 3.39, bootstrap 95% CI [1.64, 10.99]; Appendix C). The rule is exploratory and domain-conditional. It is strongest on the clinical target domain, thin-coverage on the harder multi-hop sets, and undefined for label-space answers. Its purpose is to show what the three-family decomposition makes possible as an audit policy.

The only data-dependent threshold (the per-dataset SD-UQ median) survives held-out evaluation: computing it on a random half of each dataset and applying the rule to the other half gives 91.8% mean precision (5th–95th percentile 86.7–97.4% over 200 splits) at unchanged coverage, so the in-sample number is not threshold overfitting. Per-dataset be-

Table 10: Per-dataset conjunctive audit-rule behaviour on the adaptive KG runs. Recall is the fraction of all *correct* answers selected by the rule; it is high-precision and low-recall by design, Wilson intervals expose the small denominators, and its utility presumes that unselected answers are routed to ordinary confidence handling or review rather than discarded.

Dataset	Selected	Prec.	95% CI	Recall
PubMedQA	0/100 (0%)	–	–	0.00
RealMedQA	48/223 (22%)	1.000	[0.926, 1.000]	0.23
HotpotQA	6/238 (3%)	0.833	[0.436, 0.970]	0.04
HotpotQA-FW	26/218 (12%)	0.808	[0.621, 0.915]	0.14
2WikiMHQA	4/244 (2%)	1.000	[0.510, 1.000]	0.02
MuSiQue	2/94 (2%)	0.500	[0.095, 0.905]	0.03
Pooled	86/1117 (7.7%)	0.919	[0.841, 0.960]	0.10

haviour exposes both cautionary cases: 2WikiMultiHopQA selects only four answers (2% coverage) despite weak GPS ranking, while MuSiQue selects two answers at only 50% precision. Per-domain validation is mandatory. Coverage is bounded mainly by GPS definability and SEU neutrality, so the refinements above (soft linking already; sentence-level entailment next) are the most direct route to higher audit-rule coverage. A simple learned alternative does not transfer better in this fixed-subset test: leave-one-dataset-out logistic gating selects 7.5% of answers at 85.7% precision. With three scalar features and substantial cross-dataset shift, conjunctive gating remains the more auditable policy.

8 Conclusion

Retrieval-state lock-in turns agreement into the wrong kind of reassurance. When the retriever repeatedly returns the same empty or wrong state, more samples need not test more possibilities; they may simply repeat the same condition. The confidence question therefore changes. It is not only whether the answer is stable, but which object is stable: the answer surface, the retrieved evidence, or the retrieval state itself.

The experiments show that answer-state uncertainty remains a useful default, but a bounded one. In the deployable-policy runs, 42–59% of errors already have zero observed answer dispersion at $N=5$, so an answer-only method has no within-question disagreement signal to recover. Evidence-state diagnostics can catch some of these failures by testing support in the retrieved text. GPS adds a narrower graph-support view, strongest on the calibrated clinical corpus and weak on several open-domain multi-hop suites. The field-level lesson is simple: agreement, support, and retrieval grounding are different observables, and a single confidence scalar hides that difference.

The practical recommendation is correspondingly conservative. Report answer-, evidence-, and retrieval-state signals separately, and call an answer low-risk only when all three agree: low answer uncertainty, adequate evidence support, and present graph support. In OntoGraphRAG, this conjunctive rule selects a small subset at 91.9% pooled accu-

racy against a 69.7% base rate, and reaches 100% precision under the automated judge on the clinical target domain at 22% coverage, pending human-expert confirmation before any clinical use. Unselected answers are not thereby wrong; they are simply not low-risk by this audit.

The unresolved case is presence lock-in: the system retrieves a coherent but wrong neighbourhood, and the answer is locally supported by that state. That is the setting where relation-level path faithfulness matters most: future systems should test not only whether an answer entity is reachable, but whether the path uses the relation the question actually asks for. The same diagnostic decomposition also suggests a control policy. When answer samples agree but evidence- or retrieval-state diagnostics disagree, the retriever should test a nearby alternative state, for example by masking the dominant intermediate anchor while preserving the seed entities. If the answer changes, it was anchored to that state; if it does not, the evidence was stable across the tested perturbation.

8.1 Limitations

The strict graph-only setting is a mechanism probe, not a deployable policy or a one-component ablation. It changes fusion, reranking, decomposition, and fallback together, so it demonstrates a regime in which answer-state uncertainty fails rather than estimating production prevalence. The route logs make removed fallback and vector augmentation the most plausible contributors in the strict clinical collapse, but a factorial ablation would be needed to apportion the effect.

GPS is a graph-support score in risk orientation: lower values mean stronger support for the generated answer, not greater correctness. It is strongest on the calibrated clinical corpus and weak on several open-domain multi-hop suites despite the sensitivity replay in Appendix D.8. Yes/no and other label-space answers remain outside its design. The retrieval-state family is therefore justified as an observable class (anchors, paths, subgraphs, and routes), while GPS is only one conditional scalar instantiation of that class.

SEU is also local: it tests answer-evidence consistency, not whether the model used the evidence. Its neutral plateau can make empty-context failures easier to rank than populated presence-lock-in failures, where the wrong neighbourhood may locally support the wrong answer.

The empirical scale supports the central contrasts but not fine leaderboard claims between neighbouring metrics. Correctness labels come from an automated semantic judge and survive an independent Llama-family re-judge, but human or domain-expert adjudication would be needed before treating silent errors as clinical risk. The experiments also use a single generation model (GPT-4o-mini), and fine-grained retrieval-state diversity is logged only in the dedicated diversity run. The paper should therefore be read as a diagnostic methodology and fixed-system empirical audit, not as a cross-model survey of lock-in prevalence.

The retrieval-state diagnostics are graph-native. The silent-error phenomenon is not: dense retrieval shows it

too. Extending the same decomposition to dense-RAG retrieval states, for example through citation-path or passage-link plausibility, is the most direct route to broader generality.

8.2 Ethical and Safety Considerations

Nothing in these results licenses autonomous clinical answering. The audit rule is deliberately high precision and low coverage; the unselected majority should be routed to ordinary confidence handling, broader retrieval, or human review. Lock-in also has an adversarial analogue: a corpus or query can be shaped to anchor retrieval in a plausible but wrong neighbourhood. Retrieval diversity constraints and anchor-masking controls are plausible defences, but they remain future work. Adversarial robustness lies outside this evaluation. The diagnostic decomposition is still a prerequisite for it: a failure mode that cannot be observed cannot be defended against.

References

- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hananeh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations (ICLR)*, 2024.
- Alexandra Bazarova, Andrei Volodichev, Daria Kotova, and Alexey Zaytsev. INTRYGUE: Induction-aware entropy gating for reliable RAG uncertainty estimation. *arXiv preprint arXiv:2603.21607*, 2026.
- Margarida M. Campos, António Farinhas, Chrysoula Zerva, Mário A. T. Figueiredo, and André F. T. Martins. Conformal prediction for natural language processing: A survey. *Transactions of the Association for Computational Linguistics*, 12:1619–1638, 2024. doi: 10.1162/tacl_a_00715.
- Eason Chen, Chuangji Li, Shizhuo Li, Zimo Xiao, Jionghao Lin, and Kenneth R. Koedinger. Comparing RAG and GraphRAG for page-level retrieval question answering on math textbook. *arXiv preprint arXiv:2509.16780*, 2025.
- Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, Dasha Metropolitanaky, Robert Osazuwa Ness, and Jonathan Larson. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024.
- Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. RAGAs: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158, 2024. doi: 10.18653/v1/2024.eacl-demo.16.

- URL <https://aclanthology.org/2024.eacl-demo.16/>.
- Ekaterina Fadeeva, Aleksandr Rubashevskii, Dzianis Piatrashyn, Roman Vashurin, Shehzaad Dhuliawala, Artem Shelmanov, Timothy Baldwin, Preslav Nakov, Mrinmaya Sachan, and Maxim Panov. Faithfulness-aware uncertainty quantification for fact-checking the output of retrieval augmented generation. *arXiv preprint arXiv:2505.21072*, 2025.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.
- Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Haoyu Han, Li Ma, Yu Wang, Harry Shomer, Yongjia Lei, Zhisheng Qi, Kai Guo, Zhigang Hua, Bo Long, Hui Liu, Charu C. Aggarwal, and Jiliang Tang. RAG vs. GraphRAG: A systematic evaluation and key insights. *arXiv preprint arXiv:2502.11371*, 2025.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. DeBERTa: Decoding-enhanced BERT with disentangled attention. In *International Conference on Learning Representations (ICLR)*, 2021. URL <https://openreview.net/forum?id=XPZiaotutsD>.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*, 2020.
- Yuntong Hu, Zhihan Lei, Zheng Zhang, Bo Pan, Chen Ling, and Liang Zhao. GRAG: Graph retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4145–4157, 2025. doi: 10.18653/v1/2025.findings-naacl.232. URL <https://aclanthology.org/2025.findings-naacl.232/>.
- Yue Huang, Lichao Sun, Haoran Wang, Siyuan Wu, Qihui Zhang, Yuan Li, Chujie Gao, Yixin Huang, Wenhan Lyu, Yixuan Zhang, Xiner Li, et al. TrustLLM: Trustworthiness in Large Language Models. *arXiv preprint arXiv:2401.05561*, 2024.
- Zhengbao Jiang, Frank Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 7969–7992, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.495. URL <https://aclanthology.org/2023.emnlp-main.495/>.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. In *International Conference on Learning Representations (ICLR)*, 2025.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023.
- Chorok Lee. Decomposing uncertainty in probabilistic knowledge graph embeddings: Why entity variance is not enough. *arXiv preprint arXiv:2512.22318*, 2025.
- Mufei Li, Siqi Miao, and Pan Li. Simple is effective: The roles of graphs and large language models in knowledge-graph-based retrieval-augmented generation. In *International Conference on Learning Representations (ICLR)*, 2025a.
- Xiaomin Li, Zhou Yu, Ziji Zhang, Yingying Zhuang, Swair Shah, Narayanan Sadagopan, and Anurag Beniwal. Semantic volume: Quantifying and detecting both external and internal uncertainty in LLMs. *arXiv preprint arXiv:2502.21239*, 2025b.
- Zixuan Li, Jing Xiong, Fanghua Ye, Chuanyang Zheng, Xun Wu, Jianqiao Lu, Zhongwei Wan, Xiaodan Liang, Chengming Li, Zhenan Sun, Lingpeng Kong, and Ngai Wong. UncertaintyRAG: Span-level uncertainty enhanced long-context modeling for retrieval-augmented generation. *arXiv preprint arXiv:2410.02719*, 2024.
- Huanshuo Liu, Hao Zhang, Zhijiang Guo, Jing Wang, Kuicai Dong, Xiangyang Li, Yi Quan Lee, Cong Zhang, and Yong Liu. CtrlA: Adaptive retrieval-augmented generation via inherent control. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages

- 12592–12618, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.652. URL <https://aclanthology.org/2025.findings-acl.652/>.
- Jiayu Liu, Rui Wang, Qing Zong, Yumeng Wang, Cheng Qian, Qingcheng Zeng, Tianshi Zheng, Haochen Shi, Dadi Guo, Baixuan Xu, Chunyang Li, and Yangqiu Song. NAACL: Noise-Aware verbal confidence calibration for robust large language models in RAG systems. *arXiv preprint arXiv:2601.11004*, 2026a.
- Shuyi Liu, Yuming Shang, and Xi Zhang. TruthfulRAG: Resolving factual-level conflicts in retrieval-augmented generation with knowledge graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026b.
- Linhao Luo, Yuan-Fang Li, Gholamreza Haffari, and Shirui Pan. Reasoning on graphs: Faithful and interpretable large language model reasoning. In *International Conference on Learning Representations (ICLR)*, 2024.
- Shengjie Ma, Chengjin Xu, Xuhui Jiang, Muzhi Li, Huaren Qu, Cehao Yang, Jiaxin Mao, and Jian Guo. Think-on-graph 2.0: Deep and faithful large language model reasoning with knowledge-guided retrieval augmented generation. In *International Conference on Learning Representations (ICLR)*, 2025.
- Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. Self-CheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Viktor Moskvoretskii, Maria Marina, Mikhail Salnikov, Nikolay Ivanov, Sergey Pletenev, Daria Galimzianova, Nikita Krayko, Vasily Konovalov, Irina Nikishina, and Alexander Panchenko. Adaptive retrieval without self-knowledge? bringing uncertainty back home. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6355–6384, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.319. URL <https://aclanthology.org/2025.acl-long.319/>.
- Bo Ni, Zheyuan Liu, Leyao Wang, Yongjia Lei, Yuying Zhao, Xueqi Cheng, Qingkai Zeng, Luna Dong, Yinglong Xia, Krishnaram Kenthapadi, et al. Towards trustworthy retrieval augmented generation for large language models: A survey. *arXiv preprint arXiv:2502.06872*, 2025.
- Alexander Nikitin, Jannik Kossen, Yarin Gal, and Pekka Marttinen. Kernel language entropy: Fine-grained uncertainty quantification for LLMs from semantic similarities. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- OpenAI. GPT-4o mini: Advancing cost-efficient intelligence. Technical report, OpenAI, 2024. URL <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*, 2024.
- Boci Peng, Yun Zhu, Yongchao Liu, Xiaohe Bo, Haizhou Shi, Chuntao Hong, Yan Zhang, and Siliang Tang. Graph retrieval-augmented generation: A survey. *ACM Transactions on Information Systems*, 2024.
- Laura Perez-Beltrachini and Mirella Lapata. Uncertainty quantification in retrieval augmented question answering. *Transactions on Machine Learning Research (TMLR)*, 2025.
- Jingxi Qiu, Zeyu Han, and Cheng Huang. SURE-RAG: Sufficiency and uncertainty-aware evidence verification for selective retrieval-augmented generation. *arXiv preprint arXiv:2605.03534*, 2026.
- Xin Qiu and Risto Miikkulainen. Semantic density: Uncertainty quantification for large language models through confidence measurement in semantic space. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- Jing Ren, Bowen Li, Ziqi Xu, Xikun Zhang, Haytham Fayek, and Xiaodong Li. When to trust: A causality-aware calibration framework for accurate knowledge graph retrieval-augmented generation. *arXiv preprint arXiv:2601.09241*, 2026.
- Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 338–354, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.20. URL <https://aclanthology.org/2024.naacl-long.20/>.
- Zhili Shen, Chenxin Diao, Pavlos Vougiouklis, Pascual Merita, Shriram Piramanayagam, Enting Chen, Damien Graux, Andre Melo, Ruofei Lai, Zeren Jiang, Zhongyang Li, Ye Qi, Yang Ren, Dandan Tu, and Jeff Z. Pan. GeAR: Graph-enhanced agent for retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 12049–12072, 2025. doi:

- 10.18653/v1/2025.findings-acl.624. URL <https://aclanthology.org/2025.findings-acl.624/>.
- Heydar Soudani, Evangelos Kanoulas, and Faegheh Hasibi. Why uncertainty estimation methods fall short in RAG: An axiomatic analysis. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 16596–16616, Vienna, Austria, 2025a. Association for Computational Linguistics. URL <https://aclanthology.org/2025.findings-acl.852/>.
- Heydar Soudani, Hamed Zamani, and Faegheh Hasibi. Uncertainty quantification for retrieval-augmented reasoning. *arXiv preprint arXiv:2510.11483*, 2025b.
- Weihang Su, Yichen Tang, Qingyao Ai, Zhijing Wu, and Yiqun Liu. DRAGIN: Dynamic retrieval augmented generation based on the real-time information needs of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12991–13013, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.702. URL <https://aclanthology.org/2024.acl-long.702/>.
- Jiashuo Sun, Chengjin Xu, Lumingyuan Tang, Shengjie Wang, Chen Lin, Yeyun Gong, Heung-Yeung Shum, and Jian Guo. Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph. In *International Conference on Learning Representations (ICLR)*, 2024.
- Yu Takahashi, Shun Takeuchi, Kexuan Xin, Guillaume Pelat, Yoshiaki Ikai, Junya Saito, Jonathan Vitale, Shlomo Berkovsky, and Amin Beheshti. Uncertainty-aware dynamic knowledge graphs for reliable question answering. *arXiv preprint arXiv:2601.09720*, 2026.
- Christian Tomani, Kamalika Chaudhuri, Ivan Evtimov, Daniel Cremers, and Mark Ibrahim. Uncertainty-based abstention in LLMs improves safety and reduces hallucinations. *arXiv preprint arXiv:2404.10960*, 2024.
- Harsh Trivedi, Niranjan Bauer, Tushar Khot, and Ashish Sabharwal. MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.
- Pragatheeswaran Vipulanandan, Kamal Premaratne, and Dilip Sarkar. Semantic uncertainty quantification of hallucinations in LLMs: A quantum tensor network based method. *arXiv preprint arXiv:2601.20026*, 2026.
- Sergii Voloshyn. L-RAG: Balancing context and retrieval with entropy-based lazy loading. *arXiv preprint arXiv:2601.06551*, 2026.
- Nassim Walha, Sebastian G. Gruber, Thomas Decker, Yinchong Yang, Alireza Javanmardi, Eyke Hüllermeier, and Florian Buettner. Fine-grained uncertainty decomposition in large language models: A spectral approach. *arXiv preprint arXiv:2509.22272*, 2025.
- Jonas Wallat, Maria Heuss, Maarten de Rijke, and Avishek Anand. Correctness is not faithfulness in retrieval augmented generation attributions. In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval*, pages 22–32. Association for Computing Machinery, 2025. doi: 10.1145/3731120.3744592.
- Fei Wang, Xingchen Wan, Ruoxi Sun, Jiefeng Chen, and Serkan O. Arik. Astute RAG: Overcoming imperfect retrieval augmentation and knowledge conflicts for large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *Proceedings of the Eleventh International Conference on Learning Representations (ICLR)*, 2023.
- Yu Wang, Nedim Lipka, Ruiyi Zhang, Alexa Siu, Yuying Zhao, Bo Ni, Xin Wang, Ryan Rossi, and Tyler Derr. Augmenting textual generation via topology aware retrieval. *arXiv preprint arXiv:2405.17602*, 2024. doi: 10.48550/arXiv.2405.17602.
- Junde Wu, Jiayuan Zhu, Yunli Qi, Jingkun Chen, Min Xu, Filippo Menolascina, Yueming Jin, and Vicente Grau. Medical graph RAG: Evidence-based medical large language model via graph retrieval-augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 28443–28467, 2025. doi: 10.18653/v1/2025.acl-long.1381. URL <https://aclanthology.org/2025.acl-long.1381/>.
- Kevin Wu, Eric Wu, and James Zou. ClashEval: Quantifying the tug-of-war between an LLM’s internal prior and external evidence. *arXiv preprint arXiv:2404.10198*, 2024.
- Zhishang Xiang, Chuanjie Wu, Qinggang Zhang, Shengyuan Chen, Zijin Hong, Xiao Huang, and Jinsong Su. When to use graphs in RAG: A comprehensive analysis for graph retrieval-augmented generation. *arXiv preprint arXiv:2506.05690*, 2025.
- Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. *arXiv preprint arXiv:2401.15884*, 2024.
- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings*

- of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2018.
- Zijun Yao, Weijian Qi, Liangming Pan, Shulin Cao, Linmei Hu, Liu Weichuan, Lei Hou, and Juanzi Li. SeaKR: Self-aware knowledge retrieval for adaptive retrieval augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 27022–27043, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.1312. URL <https://aclanthology.org/2025.acl-long.1312/>.
- Qinggang Zhang, Zhishang Xiang, Yilin Xiao, Le Wang, Junhui Li, Xinrun Wang, and Jinsong Su. FaithfulRAG: Fact-level conflict modeling for context-faithful retrieval-augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 21863–21882. Association for Computational Linguistics, 2025. URL <https://aclanthology.org/2025.acl-long.1062/>.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*, 2023.
- Wenqing Zheng, Dmitri Kalaev, Noah Fatsi, Daniel Barcklow, Owen Reinert, Igor Melnyk, Senthil Kumar, and C. Bayan Bruss. Revisiting RAG retrievers: An information theoretic benchmark. *arXiv preprint arXiv:2602.21553*, 2026.
- Dongzhuoran Zhou, Yuqicheng Zhu, Xiaxia Wang, Hongkuan Zhou, Yuan He, Jiaoyan Chen, Steffen Staab, and Evgeny Kharlamov. What breaks knowledge graph based RAG? benchmarking and empirical insights into reasoning under incomplete knowledge. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026.
- Xiangrong Zhu, Yuexiang Xie, Yi Liu, Yaliang Li, and Wei Hu. Knowledge graph-guided retrieval augmented generation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8912–8924, 2025a. doi: 10.18653/v1/2025.naacl-long.449. URL <https://aclanthology.org/2025.naacl-long.449/>.
- Yuqicheng Zhu, Jingcheng Wu, Yizhen Wang, Hongkuan Zhou, Jiaoyan Chen, Evgeny Kharlamov, and Steffen Staab. Certainty in uncertainty: Reasoning over uncertain knowledge graphs with statistical guarantees. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 8730–8752, 2025b. doi: 10.18653/v1/2025.emnlp-main.441. URL <https://aclanthology.org/2025.emnlp-main.441/>.
- Ilana Zimmerman, Jadin Tredup, Ethan Selfridge, and Joseph Bradley. Two-tiered encoder-based hallucination detection for retrieval-augmented generation in the wild. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 8–22, Miami, Florida, US, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-industry.2. URL <https://aclanthology.org/2024.emnlp-industry.2/>.
- Hanna Zubkova, Ji-Hoon Park, and Seong-Whan Lee. SUGAR: Leveraging contextual confidence for smarter retrieval. *arXiv preprint arXiv:2501.04899*, 2025.

A Residual-Mass Bound for Stabilised Retrieval

The collapsed-context approximation in Section 3 can be read as a simple mixture argument. Let $\alpha = p(c^\star | q)$ and define the residual context mixture

$$\pi_{\text{res}}(r) = \frac{1}{1 - \alpha} \sum_{c \in \mathcal{C}(q) \setminus \{c^\star\}} p(r | q, c) p(c | q),$$

whenever $\alpha < 1$. Then

$$p(r | q) = \alpha p(r | q, c^\star) + (1 - \alpha) \pi_{\text{res}}(r),$$

and therefore

$$\|p(r | q) - p(r | q, c^\star)\|_{\text{TV}} = (1 - \alpha) \|\pi_{\text{res}}(r) - p(r | q, c^\star)\|_{\text{TV}} \leq 1 - \alpha.$$

The bound is not used as an estimator in the experiments. It only states the condition under which the stylised approximation is meaningful: the residual routing mass must be small.

B KG Construction and System Configuration

B.1 Knowledge Graph Construction Pipeline

This section documents the full construction pipeline for reproducibility. The key design choice is that the KG is built passage-wise with chunk-level provenance, so every triple and path traces back to the text that licensed it.

Text chunking. Input documents are split into overlapping passages using a token-aware recursive splitter (tiktoken-based, with a character-based fallback when the tokenizer is unavailable) with chunk size $L = 1,500$ tokens and overlap $\delta = 200$ tokens. Each chunk receives an embedding at construction time using all-MiniLM-L6-v2 (384 dimensions) from Sentence Transformers (Reimers and Gurevych, 2019). Chunks are stored as CHUNK nodes in Neo4j with a SHA-1 content hash as identifier and a PART_OF edge to their parent DOCUMENT node.

Entity and relation extraction. Entities and typed relations are extracted from each chunk via a prompted LLM call (GPT-4o-mini; OpenAI, 2024). The LLM is prompted to return a JSON object with an `entities` array and a `relationships` array. Each entity has fields `id`, `type`, `name`, and `description`; each relation has `source`, `target`, `type`, and `description`. Two extraction modes are supported:

1. **Ontology-guided extraction.** When a domain type schema is provided, it is first parsed from JSON or OWL/RDF into a single typed internal representation. The extraction prompt then includes ontology entity types with descriptions and typed properties, together with relationship types and their domain/range/cardinality constraints. After extraction, entity types are normalised with ontology guidance via embedding-based cosine similarity against ontology class labels (threshold $\tau = 0.50$), with exact substring matching and keyword heuristics as fallback. Relationship labels are then canonicalised to the closest schema-compatible relationship type using domain/range compatibility and fuzzy matching when needed. In the biomedical and clinical experiments, types include DISEASE, TREATMENT, SYMPTOM, BIOMARKER, and ANATOMY.
2. **Open extraction.** Without a type schema, the LLM extracts entities and relations freely. Used for the open-domain datasets (HotpotQA, HotpotQA FullWiki, 2WikiMultiHopQA, and MuSiQue).

The important point is that the current system does *not* induce or generate an ontology from the document collection. It loads a user-provided ontology/schema and uses it to guide extraction, typing, and relationship normalisation during KG construction.

Neo4j schema. Entities are stored as ENTITY nodes with compound labels `__Entity__:Type` (e.g., `__Entity__:Disease`). Each node stores an embedding vector computed from the entity name and description. Entities are linked to originating chunks via MENTIONED_IN edges. Extracted relation types (e.g., TREATS, CAUSES, HAS_COMPLICATION) are stored as typed edges between entity nodes. Three indexes are maintained: a vector index over chunk embeddings, a vector index over entity embeddings, and a full-text keyword index over chunk content.

Table 11: System configuration for all experiments.

Parameter	Vanilla RAG	KG-RAG
Chunk size (tokens)	1,500	1,500
Chunk overlap (tokens)	200	200
Embedding model		all-MiniLM-L6-v2 (384d)
Top- k chunks	10	10
Adjacent chunk expansion	yes (pos ± 1)	–
Similarity threshold	0.10	0.10
Entity matching	–	mention ANN (≥ 0.72) when query entities are extracted; otherwise query ANN (≥ 0.55); plus exact/synonym matching
Chunk sufficiency threshold	–	2 chunks local PPR over the retrieved entity subgraph;
Graph scoring	–	fallback to hop prior when no graph edges are available
Graph expansion hops	–	2 (4 for MuSiQue)
Iterative local hop cap	–	$\min(h, 3)$ per sub-question
Neighbours per seed entity	–	30
Max seed entities in prompt	–	15
Max neighbour entities	–	10
Max direct relationships	–	25
Iterative decomposition	–	enabled when the hop target is $h \geq 2$
Fallback cascade	vector	entity-first $\rightarrow$ graph expansion $\rightarrow$ vector $\rightarrow$ text-keyword
LLM (generation)		GPT-4o-mini
Samples per question (N)		5

KG scope. Each dataset’s KG is stored as a named graph using a `kgName` attribute on all nodes, so that multiple datasets coexist in the same Neo4j instance without interference. The experiment pipeline supports two build-scope modes selected by `--dataset-kg-scope`: **evaluation_subset** (default) builds the KG from the same question slice being evaluated, which minimises KG size and keeps retrieval and knowledge in the same closed corpus; **full_dataset** builds from all available passages in the normalised dataset before evaluating the requested subset. Unless explicitly noted as a diagnostic, the reported fixed-snapshot runs use `evaluation_subset` so that the KG and dense retriever see the same closed corpus slice. This improves experimental control, but it also suppresses some of the graph-incompleteness and corpus-drift failures that would matter more in a deployment-scale setting.

Relationship verification. After extraction, each typed relationship is verified against the chunks that contributed at least one of its two endpoint entities (tracked via provenance positions stored at build time). Verification NLI is therefore run only over the source chunks, not over the full corpus; this reduces false-positive edges from entity-name collisions across unrelated passages.

Builder profiles and graph repair. The experiment runner exposes three KG-builder profiles: `full`, `lightweight`, and `auto`. Full-metrics runs use the `full` profile unless explicitly stated otherwise. That profile enables anchor-constrained extraction, self-reflection over missing entities, anchor-coverage supplementation, cross-passage relation recovery, and low-confidence triple reverification. It also enables soft entity linking, fragmentation repair, optional graph summaries, and claim extraction. For biomedical and clinical datasets the builder additionally attempts UMLS-backed normalisation when the local SciSpaCy/UMLS stack is available; if that stack is unavailable, construction continues without UMLS metadata rather than failing the run. The `lightweight` profile disables the expensive anchor, reflection, and cross-passage extras and is reserved for quick accuracy-only sweeps. The `auto` profile selects `lightweight` only for those cheap question-scoped multi-hop sweeps; otherwise it resolves to `full`.

Soft linking and fragmentation repair are deliberately conservative. The builder first canonicalises obvious aliases and near-duplicates, then adds bridge records only for high-similarity fragments rather than freely rewiring the graph. Claim extraction and graph summaries are stored as additional KG artefacts; they enrich global structure and inspection, but the eight headline uncertainty measures reported in the main paper do not depend on them as separate scoring heads.

B.2 System Configuration

Table 11 summarises the hyperparameters used across all experiments.

Configuration note. The $k = 10$ and similarity-threshold 0.10 entries in Table 11 are the settings used by all reported runs. The reported retrieval-study profile is `final_pair`: vanilla RAG is run only with the `dense_floor` configuration, while KG-RAG is run only with the `kg_entity_first` configuration. This avoids the earlier wasteful four-way crossing of dense and graph retrieval variants. The strict RealMedQA mechanism ablation uses the separate `strict_entity` profile, which disables query fusion, late interaction, dense fallback, vector augmentation, decomposition, and the runtime guardrail for KG-RAG. Iterative decomposition is also one source of retrieval variability, so Equation (3) in the main text is a stylised limit, not a literal claim that every KG-RAG call is deterministic. In practice, the stabilised regime is strongest on questions whose decomposition and entity anchors converge to the same graph neighbourhood across repeated calls. The released manifests record the corresponding stack flags for reranking, fallback, and query fusion, but this paper does not sweep k , similarity threshold, or routing flags as independent factors.

C Implementation Details and Reproducibility

Question sampling. The codebase supports both fixed-size pilot slices and larger dataset sweeps, and the main aggregate tables report the current fixed reported snapshot. PubMedQA and MuSiQue retain the original fixed $n = 100$ subsets. RealMedQA is reported on its full $n = 230$ evaluable question set because it is small enough to run in full and is the main shared-corpus clinical diagnostic. HotpotQA, HotpotQA FullWiki, and 2WikiMultiHopQA use fixed $n = 250$ subsets selected with the same seed to reduce denominator volatility in the multi-hop analyses. Clean-accuracy analyses can still have smaller effective denominators after excluding provider-side generation failures, and those answered counts are reported in the run artefacts and summarised in the main text.

For benchmark-style datasets, the current pipeline now uses deterministic subset sampling from the full evaluable pool. Given a dataset, sample size n , and subset seed s , it draws a fixed set of question IDs, persists that subset on disk, and reuses the same IDs on later runs unless a new subset is explicitly requested. This keeps KG construction and evaluation aligned for the same dataset/seed/sample-size combination. Exact question IDs are stored in the run artifacts and subset manifests.

KG construction LLM. Reported KG builds use GPT-4o-mini through the configured API provider (OpenRouter in the latest local runs) as the extraction model. Extraction is temperature-locked at $T = 0.0$ even when answer sampling uses a higher temperature. Biomedical and clinical datasets (PubMedQA, RealMedQA) use ontology-guided extraction with a domain type schema (Disease, Treatment, Symptom, Biomarker, Anatomy). Open-domain datasets (HotpotQA, HotpotQA FullWiki, 2WikiMultiHopQA, and MuSiQue) use unconstrained open extraction.

Training and fine-tuning configuration. No task-specific model training, fine-tuning, or gradient-based calibration is performed. The generation model, judge model, NLI models, and embedding model are all used as frozen off-the-shelf components. Unless a run explicitly sets `--judge-provider` or `--judge-model`, the correctness judge uses the same configured GPT-4o-mini backend as answer generation. The only learned or constructed object that changes across experiments is the dataset-specific KG produced by the extraction pipeline. Uncertainty samples are generated at $T = 1.0$ with $N = 5$ responses per question; deterministic accuracy-generation and KG-extraction calls use $T = 0.0$. Final-stage retrieval selection uses retrieval temperature 0.0 in the reported runs, with a shortlist factor of 4 retained in the code for stochastic retrieval sweeps.

NLI model. Discrete semantic entropy and the P(True)-style proxy cluster responses using bidirectional NLI entailment. All NLI inferences for DSE and the P(True)-style proxy use `microsoft/deberta-large-mnli` (He et al., 2021), run locally. SelfCheckGPT uses `roberta-large-mnli` for pairwise contradiction scoring. Entailment decisions are label-based (argmax over the three-class output): responses are placed in the same cluster when at least one direction predicts entailment and neither direction predicts contradiction.

Embeddings for geometric metrics. VN-Entropy, SD-UQ, and SRE-UQ compute sentence embeddings of sampled responses using `all-MiniLM-L6-v2` (384 dimensions), the same model used for chunk and entity indexing. Response embeddings are computed at evaluation time and are not cached.

Metric hyperparameters. GPS: maximum path length $h = 2$ hops (4 for MuSiQue), matching the retrieval hop depth; soft answer-entity linking with `all-MiniLM-L6-v2` cosine threshold $\tau = 0.60$ and distance decay $\gamma = 0.4$, both selected on the RealMedQA development run over a small grid and applied frozen to all other datasets. Reported GPS values come from a post-hoc replay on the saved answer logs and the persistent dataset KGs; no generation, retrieval, entity linking, or

Table 12: Summary of reported run artifacts: dataset, subset seed, sample size, evaluation mode, and KG build scope for each reported snapshot. All runs use GPT-4o-mini as the generation model and $N = 5$ uncertainty samples.

Dataset	Subset seed	n	Evaluation mode	KG build scope
PubMedQA	42	100	full_metrics	evaluation_subset
RealMedQA (adaptive)	42	230	full_metrics	evaluation_subset
RealMedQA (strict)	42	230	full_metrics	evaluation_subset
HotpotQA	42	250	full_metrics	evaluation_subset
HotpotQA FullWiki	42	250	full_metrics	evaluation_subset
2WikiMultiHopQA (adaptive)	42	250	full_metrics	evaluation_subset
2WikiMultiHopQA (strict)	42	250	full_metrics	evaluation_subset
MuSiQue	42	100	full_metrics	evaluation_subset

Table 13: Answered and failed rows in the reported saved runs. Failures are provider- or pipeline-side generation failures and are excluded from clean accuracy but retained in raw accounting. The RealMedQA adaptive–strict comparison is analysed on the 196 questions answered by both KG policies.

Dataset	Dense ans.	Dense fail	KG ans.	KG fail	Strict ans.	Strict fail
PubMedQA	100/100	0	100/100	0	–	–
RealMedQA	219/230	11	223/230	7	200/230	30
HotpotQA	226/250	24	238/250	12	–	–
HotpotQA FullWiki	208/250	42	218/250	32	–	–
2WikiMHQA	212/250	38	244/250	6	245/250	5
MuSiQue	90/100	10	94/100	6	–	–

LLM call was rerun. The run-directory replay artefacts store the linked entities and path lengths, and the final depth-matched GPS scoring pass is recorded under `results/analyses/`. SD-UQ: $k_{\max} = \min(N-1, 8)$ principal components (effective $k_{\max} = 4$ for the reported $N = 5$ samples), $\varepsilon = 10^{-12}$.

Compute environment. All experiments were run on a MacBook Pro with an Apple M4 chip under macOS 15.5 (ARM64), 14 CPU cores and 52 GB RAM, with NLI and embedding computations on CPU. Because KG extraction and generation use GPT-4o-mini through an API, end-to-end wall-clock time is dominated by remote model calls: roughly 8–12 hours for the original core-suite pass, with the larger shared-corpus and ablation runs longer (the HotpotQA FullWiki $n = 250$ stress run took about a day). These timings are specific to the laptop-and-API setup and would likely improve with local GPU-backed serving.

Effective denominators and bootstrap intervals. The main text reports point estimates because the analysis is organised around family-level behaviour and qualitative traces rather than around hypothesis tests over small AUROC gaps. To make the scale of the evidence explicit, the appendix reports the effective answered counts after provider-failure filtering, the usable GPS denominators after fallback abstention filtering, and percentile-bootstrap 95% intervals for the headline AUROCs. All intervals below use $B = 2000$ bootstrap resamples over answered questions within each dataset and system. For HotpotQA and 2WikiMultiHopQA in particular, the remaining GPS denominators are small enough to make the corresponding AUROCs descriptive.

Table 15 gives a descriptive reliability view of the same GPS scores. The clinical calibration domain has the expected monotone shape, while several open-domain rows do not; this is why the paper treats GPS as a conditional graph-support diagnostic rather than a calibrated probability.

Depth-matched GPS scoring. GPS uses a depth-matched decay, $\gamma^{|L_e - \hat{L}|}$, where $\hat{L}$ is the expected reasoning depth: the question’s logged hop count where available (2WikiMultiHopQA, MuSiQue), the dataset’s nominal depth by construction (HotpotQA variants, $\hat{L} = 2$), and $\hat{L} = 1$ otherwise. Because $\hat{L} = 1$ on RealMedQA, the calibration domain is untouched and the frozen (τ, γ) carry over; all other rows are single-shot evaluations of the same declared definition. This is the GPS definition used in all main tables.

Table 14: Effective denominators for the current reported runs. “Van answered” and “KG answered” count questions remaining after generation-failure filtering. “GPS usable” counts rows remaining after GPS abstention filtering. PubMedQA and MuSiQue use fixed $n = 100$ snapshots; RealMedQA uses its full $n = 230$ set; HotpotQA, HotpotQA FullWiki, and 2WikiMultiHopQA use fixed $n = 250$ snapshots. The HotpotQA FullWiki stress snapshot is also expanded in Table 21. GPS columns use the final soft-linked, depth-matched estimator replayed on the saved runs (Table 22 reports the corresponding coverage before/after); RealMedQA is the GPS calibration domain. Retrieval-state AUROC is only shown where the resulting denominator leaves an interpretable ranking problem.

Dataset	Van answered	KG answered	GPS usable	GPS AUROC [95% CI]
PubMedQA	100	100	–	–
RealMedQA	219	223	195	0.76 [0.59, 0.90]
HotpotQA	226	238	158	0.51 [0.42, 0.61]
HotpotQA FullWiki	208	218	185	0.54 [0.45, 0.63]
2WikiMHQA	212	244	182	0.38 [0.31, 0.47]
MuSiQue	90	94	83	0.68 [0.56, 0.80]

Table 15: GPS abstention and fixed-bin empirical error rates on adaptive KG runs. The three right columns report wrong/total rates within GPS-risk bins; lower GPS means stronger local graph support. This is a descriptive reliability check, not an ECE calculation, because GPS is a graph-support score rather than a calibrated probability.

Dataset	Answered	Abstained	Usable	GPS < .33	.33–.67	GPS > .67
PubMedQA	100	100 (100.0%)	0	–	–	–
RealMedQA	223	28 (12.6%)	195	0.0% (0/39)	4.4% (5/114)	14.3% (6/42)
HotpotQA	238	80 (33.6%)	158	44.4% (4/9)	38.1% (16/42)	33.6% (36/107)
HotpotQA FullWiki	218	33 (15.1%)	185	30.0% (21/70)	30.8% (16/52)	36.5% (23/63)
2WikiMHQA	244	62 (25.4%)	182	33.3% (6/18)	50.0% (10/20)	27.8% (40/144)
MuSiQue	94	11 (11.7%)	83	33.3% (3/9)	50.0% (10/20)	70.4% (38/54)

D Extended Results and Robustness Analyses

This section collects secondary analyses referenced from Section 7: silent-error threshold sensitivity, the archived concentration–dispersion traces, the within-adaptive stability check, the verbalised-confidence baseline, family disagreement, and the simple composite audit score.

D.1 Silent-Error Threshold Sensitivity

The exact-floor silent-error definition used in Table 7 is conservative: requiring only unanimous sampled answers ($DSE = 0$, any SD-UQ) raises the pooled rates to 52%/67%/88% (adaptive KG / dense / strict), and admitting one dissenting sample out of five raises them to 69%/84%/92% (Table 18), so the headline definition understates rather than inflates the blind spot. Unanimity at $N = 5$ is already informative: if the modal answer held only 0.7 probability, five identical samples would occur with probability $0.7^5 \approx 0.17$. Two 20-sample probes confirm the mass is not a budget artefact in the analysed snapshots. On the strict clinical subset, all 24 empty-route wrong answers stay perfectly unanimous at $N=20$ ($DSE = 0$ for every row; SD-UQ AUROC 0.17, unchanged from $N=5$). On *dense* 2WikiMultiHopQA (a pure non-graph retriever), 68% of wrong answers remain strictly silent and 86% stay within one dissenting sample of twenty, against 79% strictly silent at $N=5$ on the headline run (the $N=20$ probe uses an $n=100$ subset and the $N=5$ headline an $n=250$ subset, so the level at $N=20$ is the robust statistic rather than the precise $79 \rightarrow 68\%$ delta). Quadrupling the budget therefore barely moves the silent rate, including on a dense system with no graph state. These probes are deliberately narrow: they support persistence of the headline failure mode, not a full sampling-budget sweep across every adaptive and strict KG setting.

D.2 Interventional Dose-Response

The strict-clinical collapse summarised in Section 7.4 is not a chunk-level stochasticity artefact. An interventional dose-response holds the graph, policy, prompts, and questions fixed ($n = 100$, seed 42) and varies only final-stage retrieval determinism, sampling the final $k=10$ chunks from a $4k$ shortlist at retrieval temperature $\{0, 0.5, 1.0\}$. The manipulation works: mean within-question chunk overlap falls from 0.67 to 0.29, a 57% reduction. Yet the collapse does not move: SD-UQ AUROC is 0.18/0.18/0.16 across doses and the silent-error count is identical (24 of 29–30 wrong, with the same 24/24 empty-retrieval rows in each arm). Chunk-level stochasticity therefore does not dislodge the lock-in; the route decomposition

Table 16: Mean pairwise retrieval-overlap summaries from the saved reported runs. Overlap is the mean Jaccard similarity of retrieved chunk-text sets across the $N = 5$ uncertainty-sampling calls for each question, averaged over the run. The current artefacts log chunk overlap at run level; they do not yet retain per-question entity-overlap or path-overlap fields. The strict graph-only rows are mechanism stress tests, not deployment policies.

Dataset	Dense	Adaptive KG	Strict KG	Reading
PubMedQA	1.000	0.994	–	Single-abstract control; both retrievers are nearly deterministic.
RealMedQA	0.952	0.558	0.661	Strict KG concentrates the graph context relative to adaptive KG, but dense retrieval is also highly stable.
HotpotQA	0.904	0.677	–	Both systems retrieve stable chunk sets; the KG trace adds provenance but not an accuracy win.
HotpotQA FullWiki	0.832	0.516	–	The shared-corpus stress run introduces more KG route and fallback variation.
2WikiMHQA	0.840	0.802	0.540	Adaptive KG is close to dense at chunk level; strict graph-only failures are more visible in hop-wise output collapse than in aggregate chunk overlap.
MuSiQue	0.900	0.684	–	Deep chains are stable enough for audit, but early wrong anchors still damage answer quality.

in Section 7.4 locates the silent mass in the empty-retrieval state instead.

D.3 Archived Concentration–Dispersion Traces

Direct per-question evidence comes from three archived traces that retained per-question retrieval overlap (2WikiMultiHopQA $n = 100$, BioASQ $n = 57$, MuSiQue $n = 66$; Figure 7). Among *wrong* KG answers the pooled coupling is negative (Spearman $\rho = -0.27$, $p = 0.006$, $n = 101$): the more concentrated the retrieval state, the lower the wrong-answer dispersion. The coupling is carried by 2WikiMultiHopQA ($\rho = -0.32$, $p = 0.01$, $n = 60$), the bridge-style dataset where lock-in is sharpest. BioASQ’s archived overlap is saturated at 1.0, and MuSiQue’s small wrong-answer set trends the other way ($\rho = +0.25$, $n = 22$, n.s.). On the dense side the coupling is absent where overlap is near-deterministic and positive on MuSiQue ($\rho = +0.55$), nothing like the KG-side bridge pattern. These archived traces support, but do not identify, the mechanism.

D.4 Within-Adaptive Stability Check

An archived 2WikiMultiHopQA $n = 100$ trace, unlike the newer $n = 250$ run, retained per-question chunk-overlap fields; Table 25 uses it for a small within-adaptive sanity check. The high-overlap band is not a monotone proof of collapse, but a useful warning case: accuracy falls to 0.294, DSE AUROC is near chance at 0.467, and VN-Entropy AUROC falls to 0.583. SD-UQ still carries moderate signal (0.633), so the result should be read carefully. High-stability strata thus exist inside an adaptive policy, but the logger must persist per-question chunk, entity, and path overlap before this can be made a headline claim.

D.5 Dedicated Per-Sample Graph-State Diversity Run

Table 9 in the main text comes from a run built solely to close the logging gap noted above: the archived traces persisted chunk overlap alone, so the coupling could only ever be a warning case. The diversity run was executed on HotpotQA-FullWiki ($n = 221$ answered questions with complete per-sample traces, 74 of them wrong), reusing the existing FullWiki knowledge graph with no rebuild, and recording for every question the mean pairwise Jaccard overlap across the $N = 5$ samples for four retrieval-state families, namely the matched seed entities, the traversed paths, the assembled subgraph, and the retrieved chunks, together with the per-question SD-UQ used for the coupling test. The two halves it confirms, a near-stable retrieval state across samples (the premise Equation (3) assumes) and a negative overlap–dispersion coupling among wrong answers, are reported in the main text. The seed-entity coupling there is null only because seed overlap saturates near 1.0, leaving almost no variance to correlate against. The sign and magnitude of the path, subgraph, and chunk couplings

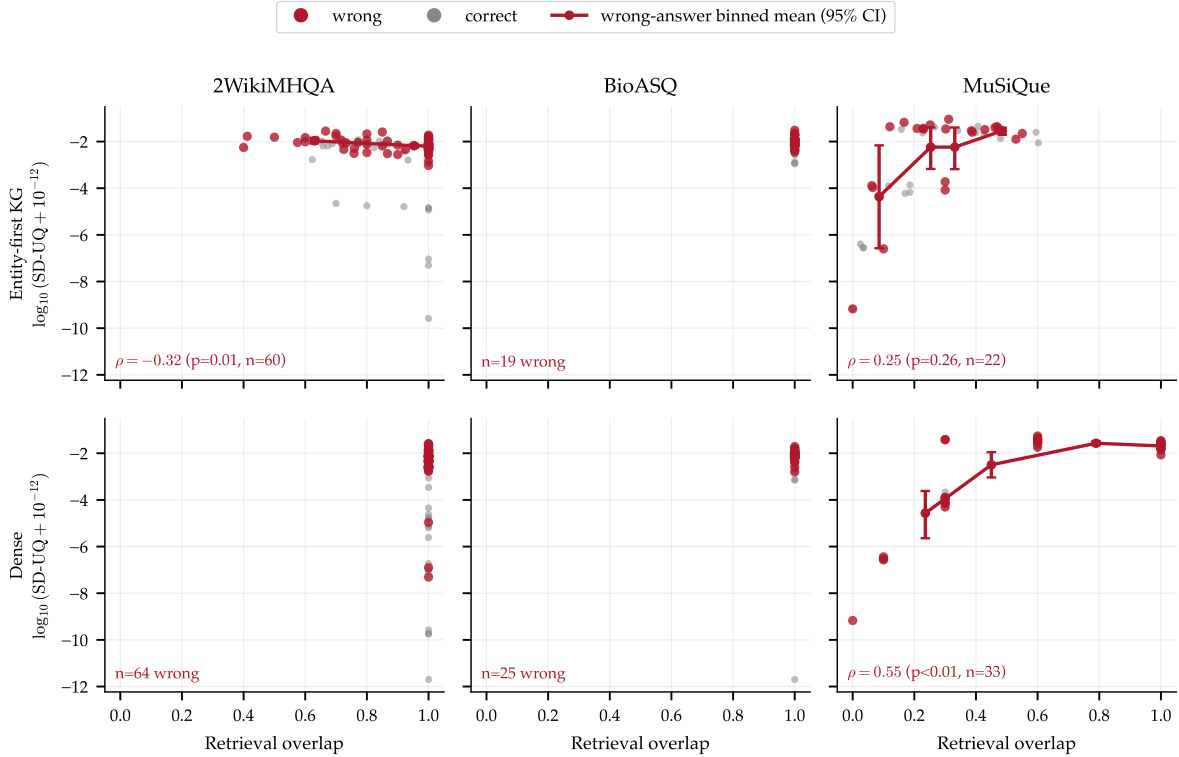


Figure 7: Full per-dataset, per-system view of the archived concentration–dispersion traces, with per-dataset Spearman correlations among wrong answers and binned means with bootstrap 95% CIs. The KG-side coupling is carried by the bridge-style 2WikiMHQA trace; BioASQ overlap is saturated at 1.0 (no within-dataset variation to correlate), and MuSiQue’s small wrong set trends positive. These details keep the pooled $\rho = -0.27$ from being read as a between-dataset artefact or as a universal law.

match the archived 2WikiMultiHopQA trace of Section D.3 on independent data and across the full set of overlap families, so the coupling no longer rests on chunk overlap from a single archived run.

D.6 Verbalised-Confidence Baseline

The verbalised-confidence baseline summarised in Section 7.3 uses the original P(True) protocol, which asks the model for a probability rather than counting cluster agreement and is therefore not blind to silent errors by construction. Querying the same backbone once per saved answer (Table 19) yields pooled adaptive-KG AUROC 0.69: competitive on the HotpotQA variants (0.72/0.77, above SD-UQ there), weaker than SD-UQ on RealMedQA (0.65 vs. 0.81), and near chance precisely where bridge-style lock-in lives (2WikiMultiHopQA: 0.49 adaptive, 0.58 strict). On the strict clinical ablation it reaches 0.89 while sampled dispersion collapses to 0.233, so part of the clinical silent-error mass is recoverable from the generator’s own self-estimate without any retrieval-side signal. If a one-call self-estimate can flag clinical silent errors, why decompose at all? First, it fails on bridge-style lock-in (0.49–0.58), the regime the decomposition targets: where the wrong answer is parametrically plausible, *no* output-side signal tested here, sampled or verbalised, ranks the errors. Second, a verbalised probability is an uncalibrated self-report with known self-preference pathologies, whereas SEU and GPS are grounded in visible evidence and graph state. Third, the self-estimate returns a number without a reason: when it disagrees with the answer there is no provenance to audit, which is precisely the evidence a clinical review workflow would need.

D.7 Family Disagreement and the Simple Composite Audit Score

Beyond the conjunctive audit rule of Section 7.8, Table 26 gives a deliberately simple audit score: the mean of within-dataset percentile ranks for SD-UQ, SEU, and final GPS-risk, with GPS abstention treated as high risk. It is uneven and not claimed to dominate the best answer-state metric. The PubMedQA row shows where the combination earns its keep: GPS abstains on every yes/no row (flat 0.5), yet Combined reaches 0.813 against SD-UQ’s 0.699, because SEU contributes complementary support-level ranking exactly where answer dispersion is coarse, not because the abstaining family adds signal. Its purpose

Table 17: Selective risk at fixed coverage on the KG side: error rate among the accepted questions when the most uncertain 20% or 10% are abstained, using a single uncertainty signal as the gate. “Base” is the no-abstention error rate. Under the adaptive policy, gating on SD-UQ alone is the strongest simple policy; SEU gating is mediocre there because 25–74% of adaptive rows sit at the all-neutral SEU = 0.5 plateau, so the 80% threshold falls inside a tie mass (adaptive SEU cells are means over 200 randomised tie-breaking draws; the spread across draws is at most 0.006). Under the strict clinical ablation the ordering reverses: SD-UQ gating is no better than accepting everything, while SEU gating removes roughly one in five residual errors at 80% coverage. The 2WikiMultiHopQA strict run is the regime where no scalar gate helps, consistent with the dataset-dependent survival reported in Section 7.5.

Run	n	Base err.	SD@90%	SD@80%	SEU@80%	Combined@80%
PubMedQA adaptive	100	0.270	0.222	0.175	0.243	0.200
RealMedQA adaptive	223	0.054	0.030	0.022	0.054	0.034
HotpotQA adaptive	238	0.395	0.393	0.384	0.405	0.395
HotpotQA FullWiki adaptive	218	0.335	0.296	0.282	0.312	0.293
2WikiMHQA adaptive	244	0.311	0.282	0.256	0.261	0.267
MuSiQue adaptive	94	0.606	0.600	0.573	0.547	0.533
RealMedQA strict	200	0.330	0.328	0.338	0.269	0.300
2WikiMHQA strict	245	0.592	0.600	0.633	0.668	0.617

Table 18: Silent-failure sensitivity ladder: pooled fraction of wrong answers classified as silent under progressively relaxed definitions of answer-state dispersion. With $N = 5$ samples DSE is quantised, so $DSE \leq 0.51$ admits at most one dissenting sample (a 4-of-5 agreement gives $DSE = 0.5004$). The headline definition in Table 7 is the most conservative rung.

Silent-error definition	Dense	Adaptive KG	Strict KG
DSE = 0 and SD-UQ at floor (headline)	0.59	0.42	0.84
DSE = 0, any SD-UQ (unanimous)	0.67	0.52	0.88
DSE ≤ 0.51 (≤ 1 dissenting sample)	0.84	0.69	0.92

is to show how the reported families can feed a decision layer; a deployable policy would still need held-out calibration.

D.8 GPS Hyperparameter Sensitivity

To test whether the headline GPS AUROCs depend on the two calibrated hyperparameters, the soft-link threshold τ and the distance decay γ are swept over $\tau \in \{0.50, 0.55, 0.60, 0.65, 0.70\}$ and $\gamma \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ (25 cells), recomputed by offline replay from the saved GPS replay stores. No generation, retrieval, linking, or KG build is rerun. Table 27 reports the calibrated cell (which reproduces the headline numbers), the AUROC range across the grid, its standard deviation, and the usable-row range. The held-out open-domain AUROCs are robust to the hyperparameters (standard deviation ≤ 0.018 on HotpotQA, HotpotQA FullWiki, 2WikiMultiHopQA, and MuSiQue), so the open-domain weakness, including the below-chance 2WikiMultiHopQA score, is a genuine retrieval limitation rather than a tuning artefact. The clinical calibration domain is the most sensitive (0.038 adaptive, 0.091 strict), as expected for the dataset the hyperparameters were tuned on.

D.9 Audit of the Fully Unflagged Residue

Of the 141 pooled adaptive-KG silent errors, 7 (5%) are unflagged by every non-answer family (Section 7.4): not on an empty route, $SEU \leq 0.5$, and GPS defined and ≤ 0.5 . A manual audit of all seven from the saved logs (question, gold answer, generated answer) characterises them as two benchmark-ambiguous or contested-label questions (e.g. the founding year of an institution with a disputed predecessor date; a subjective “more known for” comparison), two wrong-answer-slot errors where the model returns a question anchor rather than the asked attribute (e.g. naming a song’s performer instead of their cause of death), and three parametric confluences or specificity mismatches (e.g. returning a film character’s love interest instead of the lead actress’s real spouse). SEU sits at its neutral 0.5 default on four of the seven and below it on the other three, so it abstains or mildly supports rather than flagging by construction none exceed 0.5. Crucially, retrieved chunk texts are not archived, so none can be confirmed as deep presence lock-in whose evidence *entails* the wrong answer; the residue is therefore better read as a mixture of label ambiguity, answer-slot errors, and parametric overconfidence than as a wall of undetectable lock-in.

Table 19: Verbalised-confidence baseline (original P(True) protocol): the generator backbone is asked once per saved answer for the probability that the answer is correct, and $1 - p$ is scored as uncertainty against the paper’s correctness labels. No retrieval or generation is rerun. AUROC treats incorrect answers as the positive class; SD-UQ columns repeat the sampled-dispersion values for comparison. The probe is competitive on the HotpotQA variants, weaker than SD-UQ on RealMedQA, near chance on bridge-style 2WikiMultiHopQA under both policies, and, unlike sampled dispersion, survives the strict clinical ablation.

Run	Verbalised AUROC	SD-UQ AUROC
PubMedQA adaptive	0.55	0.70
RealMedQA adaptive	0.65	0.81
HotpotQA adaptive	0.72	0.64
HotpotQA FullWiki adaptive	0.77	0.67
2WikiMHQA adaptive	0.49	0.69
MuSiQue adaptive	0.65	0.63
RealMedQA strict	0.89	0.23
2WikiMHQA strict	0.58	0.52
Pooled adaptive KG	0.69	–

Table 20: Mean marginal compute time per question for each diagnostic on the KG side, averaged over the six headline runs (MacBook Pro M4; NLI on CPU). Times are *marginal*: they exclude the shared $N = 5$ answer-sampling calls that every answer-state estimator requires (the dominant latency and token cost) and the cached NLI clustering shared by the entropy-family scores. GPS varies with graph depth (its maximum, 12.3 s, is the $h = 4$ MuSiQue configuration; the clinical graph costs 0.06 s).

Diagnostic	Extra inputs	Mean time (s)
DSE / P(True) proxy	cached NLI clusters	$< 10^{-4}$
SelfCheckGPT	$O(N^2)$ NLI calls	1.68
SRE-UQ	embeddings	0.040
VN-Entropy	embeddings	0.007
SD-UQ	embeddings	0.012
SEU	K NLI calls	1.65
GPS	graph queries	2.43 (0.06–12.3)

D.10 Relocated Diagnostic Tables

Two diagnostic tables referenced from the main text are collected here to keep the results flow tight. Table 28 summarises, per family, the regime each diagnostic detects and where it goes silent. Table 29 gives the full hop-stratified 2WikiMultiHopQA comparison across the three retrieval policies; its decisive row, the strict graph-only 4-hop slice with zero clean accuracy and collapsed SD-UQ and VN-Entropy, is discussed in Section 7.4. The table uses clean answered rows rather than raw hop-bucket sizes, so n differs across policies when provider failures or skipped systems remove rows, the same convention used for clean accuracy.

E Additional Qualitative Cases

Table 30 is the canonical case gallery. The failure rows follow the form wrong anchor $\rightarrow$ stable retrieved context $\rightarrow$ stable wrong answer $\rightarrow$ silent answer surface, showing per case which diagnostic family still responds; the success rows show the same retrieval stability working as intended on 2WikiMHQA bridge questions, where the graph preserves the intermediate entity that vanilla retrieval loses, plus one MuSiQue chain resolved correctly despite GPS abstention. The main text uses one path-faithfulness example (the Osireion lock-in of Figure 1) to keep the exposition focused on the lock-in mechanism; the gallery provides trace evidence, not additional leaderboard claims. Table 31 gives the full per-family audit readout for that Osireion case (question, retrieved graph state, and the DSE/SD-UQ/GPS/SEU verdicts), expanding the summary in Section 7.7.

F Prompts

All prompts used in the experiments are reproduced verbatim below.

Table 21: HotpotQA FullWiki $n = 250$ shared-corpus stress run. Vanilla uses dense retrieval; KG-RAG uses the reported entity-first policy with fallback. Clean accuracy excludes provider-side generation failures. GPS AUROC is reported only for KG-RAG because GPS is a graph-state diagnostic; the GPS null rate is the fraction of answered KG rows for which the metric is unavailable or falls back.

Policy	Ans.	Clean acc.	Raw acc.	DSE	SRE-UQ	VN-Ent.	SD-UQ	SEU	GPS / null
Dense vanilla	208	0.721	0.600	0.650	0.630	0.666	0.665	0.563	–
Entity-first KG	218	0.665	0.580	0.666	0.688	0.701	0.672	0.565	0.543 / 0.151

Table 22: Post-hoc RealMedQA GPS replay on the full $n = 230$ answer logs. No generation, retrieval, or LLM call was rerun; only GPS was recomputed against the persistent dataset KG. “Old” is the original surface-matched, binary-reachability GPS; “new” is the final soft-linked, depth-matched estimator of Section 5.2. The linking threshold and decay ($\tau = 0.60$, $\gamma = 0.4$) were selected on this dataset and applied frozen everywhere else, so this table shows calibrated in-domain behaviour; the open-domain tables provide the held-out estimate.

Setting	Answered	GPS usable (old→new)	GPS AUROC (old→new)
Dense + vanilla	219	100 → 192	0.72 → 0.89
Entity-first + KG	223	97 → 195	0.47 → 0.76

F.1 Correctness Judge Prompt

Used to produce the binary correctness labels on which AUROC and AUREC are computed.

System message:

You are a strict answer evaluator for a factoid question answering task. Your job is to decide if a model's response is CORRECT.

Rules:

- Reply with exactly one word: 'correct' or 'incorrect'.
- The response is CORRECT only if it contains an answer semantically equivalent to the expected answer (minor spelling/accent differences are ok).
- The response is INCORRECT if the model says it doesn't know, cannot determine, or provides a factually different answer.

User message (template):

Question: {question}

Expected answer: {expected_answer}

Model response: {model_response}

Is the model response correct? Reply with one word only:
correct or incorrect.

F.2 Vanilla RAG Generation Prompt

Used by the vanilla RAG system to generate all $N = 5$ sampled responses per question.

System message (template):

You are an AI assistant that provides accurate, factual answers based on the provided context.

Context Information:

{context}

Guidelines:

- Read all context passages carefully -- the answer is often present but may require connecting two passages.
- For multi-hop questions: explicitly chain your reasoning step by step (e.g. "The film starred X -> X later held position Y").
- Base your answer on the provided context; do not invent facts.

Table 23: RealMedQA no-rebuild strict graph-only stress test on the full $n = 230$ question set. The adaptive row is the reported KG policy from the full RealMedQA run. The strict row reuses the same RealMedQA graph, disables dense fallback, vector augmentation, and decomposition, and therefore tests whether a more concentrated entity-first retrieval regime compresses answer-state uncertainty. All metric columns report AUROC.

KG policy	Ans.	Acc.	Overlap	DSE	VN	SD-UQ	SEU	GPS
Adaptive + fallback	223	0.946	0.558	0.699	0.815	0.808	0.501	0.76 [†]
Strict graph-only	200	0.670	0.661	0.463	0.248	0.233	0.721	0.746 [†]

[†] GPS values come from the post-hoc replay (Table 22): adaptive 195/223 usable rows, strict 142/200. RealMedQA is the calibration domain for the GPS hyperparameters, so these cells show calibrated in-domain behaviour; Table 15 gives the corresponding binned error-rate view.

Table 24: Percentile-bootstrap 95% intervals for the headline answer-state and evidence-state AUROCs. These are the main answer-side metrics used in the narrative: SRE-UQ and SD-UQ as the practical answer-state baselines, and SEU as the evidence-state signal. The table is intended to calibrate scale, not to claim significance testing for every pairwise gap.

Dataset	SRE-UQ Van.	SRE-UQ KG	SD-UQ Van.	SD-UQ KG	SEU Van.	SEU KG
PubMedQA	0.56 [0.49, 0.64]	0.64 [0.55, 0.73]	0.60 [0.46, 0.74]	0.70 [0.57, 0.81]	0.55 [0.44, 0.67]	0.59 [0.47, 0.71]
RealMedQA	0.43 [0.24, 0.62]	0.66 [0.51, 0.80]	0.63 [0.39, 0.86]	0.81 [0.64, 0.94]	0.63 [0.47, 0.79]	0.50 [0.35, 0.63]
HotpotQA	0.61 [0.54, 0.68]	0.63 [0.57, 0.70]	0.64 [0.56, 0.72]	0.64 [0.56, 0.70]	0.46 [0.39, 0.54]	0.48 [0.40, 0.55]
HotpotQA FullWiki	0.63 [0.55, 0.71]	0.69 [0.60, 0.76]	0.66 [0.58, 0.75]	0.67 [0.60, 0.75]	0.56 [0.47, 0.65]	0.56 [0.49, 0.64]
2WikiMHQA	0.59 [0.52, 0.67]	0.65 [0.58, 0.72]	0.61 [0.52, 0.69]	0.69 [0.61, 0.75]	0.62 [0.53, 0.70]	0.64 [0.57, 0.72]
MuSiQue	0.64 [0.53, 0.75]	0.66 [0.56, 0.77]	0.67 [0.55, 0.78]	0.63 [0.52, 0.75]	0.63 [0.52, 0.75]	0.63 [0.51, 0.74]

- If the answer is not directly stated but can be inferred by connecting two pieces of evidence, make the inference explicitly and state your reasoning chain.
- Only say the context is insufficient if you genuinely cannot find any relevant evidence after carefully reading all passages.
- Be concise but comprehensive; include specific facts to support your answer.
- IMPORTANT: For yes/no questions (questions starting with: Is, Are, Does, Do, Can, Should, Was, Were, Has, Have), you MUST begin your response with either "Yes" or "No" as the very first word, followed by your explanation.

User Query: {question}

F.3 KG-RAG Generation Prompt

Used by the KG-RAG system to generate all $N = 5$ sampled responses per question. The instruction to prefer evidence appearing in *both* text chunks and graph paths is the proximate cause of consistent, confident generation under KG context, one mechanism that amplifies graph-induced overconfidence.

System message (template):

You are a knowledgeable AI assistant. Answer the question using the provided context.

The context has three parts:

1. TEXT CHUNKS -- document passages retrieved by semantic similarity.
2. GRAPH TRAVERSAL PATHS -- multi-hop chains discovered by walking the knowledge graph from seed entities.
Each path shows how concepts connect:
Entity A --RELATIONSHIP--> Entity B --RELATIONSHIP--> Entity C
3. ENTITIES -- individual concepts found in the graph.

HOW TO REASON:

- Read all text chunks carefully -- the answer is often present but requires connecting two passages.
- Start from the entities most relevant to the question, then follow

Table 25: Within-adaptive retrieval-stability sanity check from an archived 2WikiMultiHopQA $n = 100$ trace that retained per-question chunk-overlap fields. The newer $n = 250$ run is the headline result; this table is a mechanism check. Bands are population tertiles of per-question chunk overlap; because a large fraction of questions sit at exactly 1.0 overlap, ties spill across the tertile boundary. AUROC treats incorrect answers as the positive class.

Overlap band	n	Range	Acc.	DSE AUC	SD AUC	VN AUC	DSE mean
Low	33	0.400–0.840	0.364	0.415	0.714	0.750	0.277
Medium	33	0.850–1.000	0.545	0.667	0.800	0.752	0.103
High	34	1.000–1.000	0.294	0.467	0.633	0.583	0.157

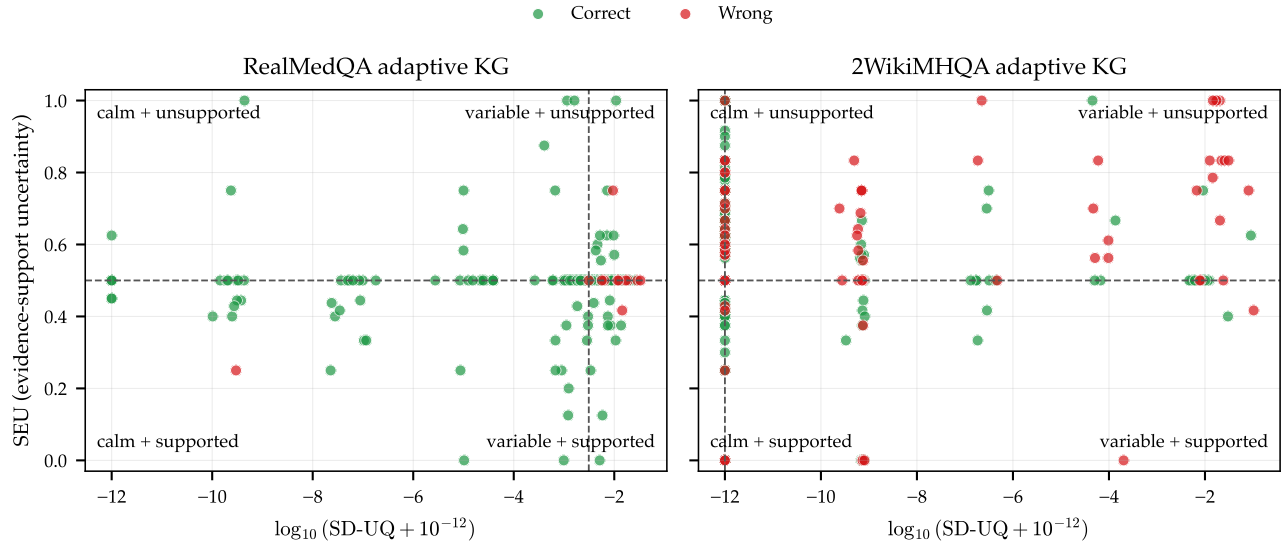


Figure 8: Family disagreement in the pooled adaptive KG runs. Grey hexagons show the density of all answered questions with a non-neutral SEU verdict; red points overlay the wrong answers. The x -axis is answer-state residual dispersion (SD-UQ, log scale) and the y -axis is evidence-state uncertainty (SEU). The all-neutral SEU = 0.5 rows, where the NLI head said nothing ($n = 464$, 24% wrong), are excluded. Dashed lines mark the pooled SD-UQ median and SEU = 0.5; the two low-answer-uncertainty quadrants carry the decision-relevant contrast: low answer uncertainty with contradicted evidence is 31% wrong against 20% for low answer uncertainty with support, so answer-surface agreement is not a proxy for evidence support.

- graph paths to discover indirect relationships.
- For multi-hop questions: explicitly chain your reasoning step by step (e.g. "The film starred X -> X later held position Y").
- Prefer evidence that appears in BOTH a text chunk AND a graph path -- that is the strongest signal.

IMPORTANT:

- For yes/no questions, begin with "Yes" or "No" followed by your explanation.
- Ground every claim in the provided context; do not invent facts.
- If the answer is not directly stated but can be inferred by connecting two pieces of evidence in the context, make the inference explicitly and state your reasoning chain.
- Only say the context is insufficient if you genuinely cannot find any relevant evidence after carefully reading all passages.

Text Chunks:

{context}

Knowledge Graph Traversal Paths:

{graph_paths}

Table 26: Adaptive-KG composite audit score. The combined score is the mean of within-dataset percentile ranks for SD-UQ, SEU, and final GPS-risk, with GPS abstention treated as high risk. AUROC treats incorrect answers as the positive class. “Base err.” is the no-abstention error rate; lower AUREC is better within a dataset. “Best single” is the largest per-family AUROC in the row.

Dataset	n	Base err.	SD-UQ AUC	SEU AUC	GPS-risk AUC	Best single	Combined AUC	Combined AUREC
PubMedQA	100	0.270	0.699	0.594	0.500	0.699	0.813	0.096
RealMedQA	223	0.054	0.808	0.501	0.703	0.808	0.729	0.022
HotpotQA	238	0.395	0.635	0.477	0.551	0.635	0.573	0.326
HotpotQA FullWiki	218	0.335	0.672	0.565	0.549	0.672	0.661	0.228
2WikiMHQA	244	0.311	0.685	0.643	0.414	0.685	0.619	0.225
MuSiQue	94	0.606	0.631	0.626	0.640	0.640	0.745	0.426

Table 27: GPS AUROC sensitivity to (τ, γ) by offline replay. “Calib.” is the paper’s frozen $\tau=0.60$, $\gamma=0.40$ cell; range, SD, and usable counts are over the 5×5 grid.

Run	Calib.	Grid range	SD	Usable
RealMedQA adaptive	0.76	0.60–0.76	0.038	191–197
RealMedQA strict	0.75	0.53–0.90	0.091	139–171
HotpotQA	0.51	0.50–0.55	0.014	154–169
HotpotQA FullWiki	0.54	0.53–0.56	0.008	182–187
2WikiMHQA adaptive	0.38	0.35–0.40	0.012	177–188
2WikiMHQA strict	0.48	0.40–0.53	0.035	94–104
MuSiQue	0.68	0.64–0.70	0.018	80–85

Entities:
{entities}

Question: {question}

F.4 Entity Extraction Prompts

The prompts used for LLM-based entity and relation extraction during KG construction (open and ontology-guided modes) follow the JSON schema described in Appendix B.1. They are available in the supplementary code release (Appendix H).

G OntoGraphRAG System Description

OntoGraphRAG (v1.0.0-rc1) is the open-source platform used to run all reported experiments. It exposes a FastAPI application on port 8004, started with `uvicorn ontographrag.api.app:app --host 0.0.0.0 --port 8004` or via the installed console script `ontograph`. Neo4j is checked at startup; endpoints requiring the graph database return HTTP 503 if unavailable.

G.1 Package Structure

`ontographrag.kg` KG construction. `builders/` contains the open extraction and ontology-guided extraction pipelines. `loaders/` handles Neo4j serialisation, and `utils/` contains chunking, source-node, and graph-query helpers.

`ontographrag.rag` EnhancedRAGSystem (KG-RAG pipeline) and VanillaRAGSystem (dense-retrieval baseline), with auxiliary modules for reranking, retrieval sampling, and answer guardrails.

`ontographrag.providers` Unified LLM interface supporting OpenAI, Anthropic, Google Gemini, Vertex AI, OpenRouter, and Ollama.

Persistence: Neo4j ≥ 5.25 for all graph, vector, and full-text indexes. Chunk and entity embeddings are stored as Neo4j native vector indexes; no external vector store is required. Install with `pip install .`; deployment requires Python ≥ 3.11 and Neo4j connection environment variables.

Retrieval regime	State of the context	Answer-state	SEU	GPS	Reading
Anchor failure	Entity linking fails; system falls back to text retrieval.	~ variation may persist	~ fallback passages	⊘ abstains	Absence of a graph object is itself diagnostic.
Stabilised retrieval	Same or near-same graph neighbourhood across samples.	↓ compressed	✓ support / contradiction	✓ consistency	Agreement should be read against evidence and graph.
Wrong coherent anchor	Anchoring succeeds on the wrong neighbourhood; paths locally coherent.	↓ same wrong answer repeats	~ can warn	× supports wrong state	Hardest lock-in case; trace inspection needed.
KG incompleteness	Plausible anchor, but bridges or relations missing from the graph.	~ fallback stores variation	✓ if text re-mains	⊘/~ reduced	Coverage or construction error, not decoder uncertainty.

Table 28: Expected diagnostic behaviour across KG-RAG retrieval regimes. Symbols: ✓ informative; ~ partially informative; ↓ compressed and potentially silent; ⊘ abstains; × locally consistent with a wrong state. Each cell states what a family can observe, not a ranking; Table 2 defines the families and Table 6 gives the resulting 2×2 taxonomy.

Table 29: Hop-wise 2WikiMultiHopQA diagnostics on the same fixed $n = 250$ subset. The table uses clean answered rows; n_w is the AUROC positive class. AUROC is omitted when a hop slice has only one correctness class. The ≥ 5 -hop bucket contains four questions and is kept only in the released JSON artefact.

Policy	Hop	n	n_w	Acc.	SD mean	VN mean	SEU mean	SD AUC	VN AUC	SEU AUC	GPS AUC
Dense vanilla	2-hop	150	42	0.720	0.000	0.092	0.612	0.615	0.639	0.615	0.491
Dense vanilla	4-hop	58	16	0.724	0.000	0.198	0.458	0.628	0.571	0.656	0.410
Adaptive KG	2-hop	182	57	0.687	0.003	0.191	0.544	0.702	0.668	0.611	0.335
Adaptive KG	4-hop	58	16	0.724	0.002	0.248	0.466	0.600	0.638	0.646	0.452
Strict KG	2-hop	183	83	0.546	0.003	0.127	0.569	0.560	0.680	0.378	0.353
Strict KG	4-hop	58	58	0.000	0.000	0.000	0.500	–	–	–	–

G.2 Serving Interface

The full REST API is documented in the accompanying software release rather than reproduced here. For the reported experiments, the relevant interface is the question answering endpoint: it returns the generated answer, retrieved text chunks, matched entities, graph paths, route labels, relation anchors, and per-query diagnostic scores. The manuscript therefore treats the API as infrastructure for trace logging, not as a separate systems contribution.

H Code and Data Availability

The system implementation, the experiment harness, and the scripts that generate every table and figure in this paper are available at <https://github.com/julka01/OntoGraphRAG>. The supplementary run artefacts are released in the same repository: the raw per-run logs, the fixed sampled question IDs for each snapshot, both judges’ correctness labels, the raw per-query metric scores and uncertainty estimates, and the GPS replay stores used for the post-hoc recomputations (the RealMedQA replay, the $\tau \times \gamma$ sensitivity sweep, and the HotpotQA-FullWiki diversity run). The release bundle is `reproducibility/arxiv_v1/`, and `reproducibility/arxiv_v1/MANIFEST.json` records the exact repository commit and file hashes for the cached logs used in the paper. Because every fixed-subset number and every post-hoc replay in the paper is computed from these saved artefacts, the reported results can be reproduced without rerunning generation or retrieval.

Reproducibility checklist. Table 32 consolidates the exact configuration values scattered through this appendix.

Mechanism-probe artefacts. The post-submission mechanism analyses reported in Section 7.4 have their own artefacts, released alongside the run logs: the retrieval-temperature dose-response runs (RealMedQA, strict profile,

Table 30: Consolidated case gallery from the reported runs. × rows are verified lock-in failures: the KG answer is wrong yet repeated across all samples (DSE = 0, SD-UQ at or near the floor); the diagnostic columns show which family, if any, still fires. ✓ rows are KG successes on bridge questions, where the same retrieval stability is helpful because the path reaches the right entity and relation, plus one correct answer despite GPS abstention. The Baltic Cup row is the hardest failure: the retrieved evidence *entails* the wrong answer, so every scalar family reports low risk and only the provenance trace remains auditable. GPS = 0.5 denotes abstention. The cases are selected for illustration, not sampled; prevalence claims rest on Tables 7 and 6, not on this gallery.

KG	Question (dataset)	Gold	Anchor / path	KG answer	SEU	GPS
×	Osireion is behind the temple of which pharaoh? (HotpotQA)	Seti I	<i>Ramesses II</i> neighbourhood of the Abydos temple	Ramesses II	1.00	0.44
×	Who is the paternal grandfather of Uskhal Khan? (2Wiki)	Khutughtu Khan	<i>Toghon Temür</i> (father, one hop; bridge skipped)	Toghon Temür	0.60	0.00
×	Who created Mickey Mouse’s spouse? (MuSiQue)	Walt Disney	<i>Ub Iwerks</i> via Mickey Mouse → creator	Ub Iwerks	0.75	0.00
×	Who is Ieuan ab Owain Glyndŵr’s paternal grandfather? (2Wiki)	Gruffudd Fychan II	<i>Owain Glyndŵr</i> (father, one hop; bridge skipped)	Owain Glyndŵr	0.43	0.24
×	Which country skipped the 1991 Baltic Cup? (HotpotQA)	Belarus	<i>Estonia</i> via Baltic Cup participants	Estonia	0.00	0.00
×	Are TEC-1 and Dubna 48K based on the same processor? (HotpotQA)	yes	weak graph state; no usable anchor, GPS abstains	no	0.75	0.50
×	2010 population of the city popular with tourists? (MuSiQue)	8.005 million	wrong city; only linkable answer string four hops away	2,217	0.67	0.94
✓	Who is the spouse of the director of <i>Fire-Eater</i> ? (2Wiki)	Pirkko Saisio	film → director → spouse bridge preserved	Pirkko Saisio	0.60	0.44
✓	Who is the father of the director of <i>The Cup</i> (1999)? (2Wiki)	Thinley Norbu	film → director → father bridge preserved	Thinley Norbu	0.69	0.35
✓	Birthplace of the director of <i>Dollar</i> (1938)? (2Wiki)	Helsingfors	film → director → birthplace bridge preserved	Helsingfors	0.50	0.60
✓	Dow Jones fall at the highest US unemployment rate? (MuSiQue)	54.7%	chain resolved correctly; GPS still abstains	54.7%	0.72	0.50

$n = 100$, seed 42, retrieval temperatures $\{0, 0.5, 1.0\}$, with per-question chunk overlap retained), the $N=20$ resampling probe on the same strict subset, the independent re-judge of all free-text runs (`rejudge_independent.json`, `rejudge_wave1.json`), the verbalised-confidence baseline (`verbalized_confidence.json`), the audit-rule evaluation with matched-coverage baselines and split-half validation (`certificate_eval.json`), the silent-rate sensitivity and route-decomposition numbers (`wave1_analysis.json`), the reporting-robustness bookkeeping for failure counts and the audit rule odds ratio (`reporting_robustness.json`), and the KG scale statistics queried from the persistent Neo4j stores (`kg_scale_stats.json`).

Table 31: Path-faithfulness failure under retrieval lock-in (HotpotQA). A wrong-but-graph-coherent case from the fixed- $n = 250$ run. The retriever anchors to a strongly associated but incorrect Nineteenth-Dynasty pharaoh, and every sampled answer repeats it. Answer-state metrics are silent and GPS reports support; only SEU fires.

Question and gold answer	Retrieved graph state	Diagnostic reading
<i>Osireion is located to the rear of the temple named after which New Kingdom Nineteenth Dynasty of Egypt pharaoh?</i> Gold answer: <i>Seti I</i>. The intended chain attributes the Great Temple of Abydos—directly in front of the Osireion—to its builder <i>Seti I</i>.	The retrieved subgraph anchors instead to <i>Ramesses II</i>, the adjacent Nineteenth-Dynasty pharaoh who completed the temple and built his own nearby: Osireion / Abydos temple ↓ associated pharaoh Ramesses II The generated KG answer is therefore <i>Ramesses II</i> (×), and dense retrieval makes the same error. The graph state is locally coherent but attached to the wrong builder.	All sampled KG answers repeat the same wrong entity: DSE = 0, SD-UQ $\approx$ 0. Answer-state uncertainty is therefore at its floor. GPS is low (0.44), so it does not flag the error: <i>Ramesses II</i> is reachable in the retrieved graph, and GPS is reported in risk orientation, where lower means stronger graph support for the generated answer. SEU is maximal (1.0): every retrieved chunk contradicts the generated answer, the one family that fires.

Table 32: Reproducibility checklist (verified against the released run artefacts and code).

Item	Value
Generation model	gpt-4o-mini via OpenRouter (openai/gpt-4o-mini); $T=1.0$ for the $N=5$ uncertainty samples, $T=0.0$ for accuracy generation and KG extraction
Judge model (main)	same gpt-4o-mini backbone as generation
Judge model (re-judge)	meta-llama/llama-3.3-70b-instruct via OpenRouter, $T=0.0$
Embedding model	all-MiniLM-L6-v2 (384-d, Sentence Transformers), shared by chunks, entities, and response embeddings
NLI models	microsoft/deberta-large-mnli (DSE / P(True) clustering, SEU); roberta-large-mnli (SelfCheckGPT pairwise)
Subset seeds	42 for every dataset and run (including dose-response and $N=20$ probes)
Sampling budget	$N=5$ headline; $N=20$ resampling probe
GPS calibration and sensitivity	$\tau=0.60$, $\gamma=0.40$ selected on the RealMedQA replay and applied frozen elsewhere; Appendix D.8 replays $\tau \in \{0.50, \dots, 0.70\}$ and $\gamma \in \{0.2, \dots, 1.0\}$
Cached reproduction	headline tables and figures regenerate from released saved artefacts; no API calls, KG rebuild, or generation rerun is required for the reported analyses
Retrieval temperature	0.0 in all reported runs; $\{0, 0.5, 1.0\}$ in the dose-response arms (shortlist factor 4)
Provider failures	per-run generation failures excluded from clean accuracy: KG side 0–32 and dense side 0–42 per run (Table 13)
Code version	OntoGraphRAG v1.0.0, public repository; exact commit and artefact hashes recorded in the released manifest